EMS slew limiters

Synthi 100 and Synthi Sequencer 256

Research packet for Teletape Workshop / Triple Slew Limiter
Compiled 13 October 2026

What is in this packet

A consolidated record of the slew-limiter specifications, control behavior, circuit evidence, engineering clues, and unresolved questions found so far. Original archive pages follow the notes at their original PDF resolution. The complete 25-page Sequencer 256 service collection is included so related signal definitions and circuit sheets remain available together.

The strongest findings

Synthi 100: three unity-gain followers with manual and exponential voltage control of maximum slew. EMS circuit drawing 32/12 Issue C is available, including the transistor network and integrating capacitor.

Sequencer 256: the operating manual gives 0.05-1000 V/s for Layer 1 Voltage A. Advancing the control restricts rate and adds portamento. The service system diagram locates a slew-limiter block after the Layer 1 A output stage.

Relationship: the user reports that the standalone rack units were derived from Synthi 100 circuits. The shared architecture is plausible, but a complete revision-by-revision match between the Synthi 100 TSL, standalone rack TSL, and 256 slew stage has not been established.

How to read the evidence

Primary evidence means an EMS manual, specification, or schematic. **Interpretation** means a circuit or DSP reading derived from that evidence. **Prototype choice** means our software behavior. Gathered forum fragments are kept as leads, with unsupported claims identified.

The internal 0.0-1.0 knob range and current CV calibration belong to the Voltage Modular prototype. They are not historical measurements.

1 / Synthi 100 specifications

Source: four-page Computer Synthi brochure, archive PDF p. 2 (packet p. 7). Its treatment-module description is the principal published performance reference.

Parameter	Published evidence / interpretation
Channels	Three voltage-controlled slew limiters.
Steady-state behavior	Unity-gain following: the output reaches the input level. EMS states that no output-level control is provided.
Gain qualification	Gain is listed as 1 with a ca. 1% qualification. The exact tolerance notation is unclear in the available transcription; consult the reproduced page.
Steady-state linearity	Approximately 0.05% BSL (best straight line). This is a steady-state figure, not ramp-curvature or slew-accuracy data.
Slew-control range	1 ms-10 s. The voltage excursion used for these times is not stated.
Manual control	A panel knob controls maximum rate of change.
External control	The maximum slew is set by control voltage; the brochure describes exponential voltage control.
Musical example	Use keyboard-touch/velocity information to change pitch-glide rate between notes.

What these numbers do not tell us

A time range alone is insufficient to derive V/s. For a constant-rate transition, time = voltage excursion / rate. A 1 V and a 5 V change therefore take different times at the same rate. The 1 ms-10 s specification cannot be equated numerically with the 256 range without an excursion or calibration procedure.

The rack photograph establishes numbered 0-10 positions, three voltage controls, voltage inputs/outputs, V.C. inputs, and a lamp. It does not establish pot taper, volts per control division, CV polarity, or CV sensitivity. A 1975 Studio Sound equipment survey (printed p. 55) corroborates three Synthi 100 limiters and the 1 ms-10 s range; it supplies no dial mapping.

2 / Synthi 100 circuit evidence

Primary sheet: EMS 32/12, Issue C, "Triple Slew Limiter"; Synthi 100 service archive PDF p. 25 (packet p. 8). One channel is drawn with alternative connector numbers for the repeated circuitry.

Block / item	Direct observation
Signal path	Voltage input, transistor/op-amp network, and SLEW OUT. A1 has a feedback capacitor; A2 senses the signal/output network.
Rate controls	Separate POT SLEW RATE (RV421) and I.C. SLEW RATE CONTROL inputs. Manual pot is drawn between +12 V and -12 V.
Supply	+12 V, -12 V and 0 V. These are supply rails, not a verified allowed signal range.
Op amps	A1 and A2 are marked SN72741P: 741-family op-amp sections.
Transistors	Q1/Q2 marked 2N4283; Q3/Q4 marked 2N5172. Q5 has a faded 2N428-series marking, previously transcribed as 2N4283; its final character deserves a cleaner scan.
Timing capacitor	C1 = 0.22 uF in A1 feedback. Dielectric/material is not specified.
Trimmers	PR1 = 1 kOhm and PR2 = 10 kOhm are visible. Their exact adjustment procedures are not provided on this sheet.
R2 revision note	Schematic body shows 33 kOhm. Margin says "R2 was 39K", dated 7 August 1971. Preserve both annotations; the note likely identifies a former value, but no full revision procedure is supplied.

Functional reading - interpretation

The capacitor, transistor-controlled current paths and feedback sensing support a controlled integrator/follower interpretation. Manual and external rate controls are distinct from the signal voltage. No separate rise/fall controls appear on the channel sheet. An exponential control law need not produce an exponential output ramp: an exponentially selected speed can still produce a straight, rate-limited transition.

The Random Generator sheet, EMS 32/08 (archive p. 21; packet p. 9), separately identifies ramp-and-hold and slew-limited output shaping. It is related EMS timing practice, not the TSL channel circuit.

3 / Sequencer 256 slew behavior

Primary operating-manual source: EMS Synthi Sequencer 256 operating manual, printed p. 7 (PDF p. 7), Output Controls. The paragraph concerns Layer 1 Voltage A only.

Its slew control spans 0.05-1000 V/s. At the minimum control setting, 1000 V/s is nearly instantaneous for the stated sequencing application. Advancing the control restricts rate and adds pitch portamento. The other voltage outputs do not have that added slew control. This verifies the 256 behavior; it does not independently calibrate the rack TSL.

Placement and conversion - archived service evidence

The 256 records sampled voltage values digitally with event timing. Service sheets identify a 6-bit ADC (archive p. 19, EMS 32/41) and DAC circuitry (archive p. 23, EMS 32/43). The system drawing, 61-27 (archive p. 25; packet p. 35), shows the Layer 1 A output stage feeding a separate SLEW LIMITER block, controlled by the Voltage A Slew Rate pot. Layer 1 B and the remaining layers follow their own output paths.

The system drawing locates the slew function but does not show its internal component circuit. The Synthi 100 EMS 32/12 sheet is the detailed slew circuit available in this collection; an exact 256-board equivalence is not demonstrated by a black-box label.

Rate-to-time examples - calculated, not additional specifications

Rate	1 V transition	5 V transition
1000 V/s	1 ms	5 ms
100 V/s	10 ms	50 ms
1 V/s	1 s	5 s
0.05 V/s	20 s	100 s

The Synthi 100 sequencer system drawing (archive p. 63; packet p. 10) is also reproduced as routing context. It is not a component-level diagram of the 256 slew stage. The entire 256 service collection appears at packet pp. 11-35.

4 / Implications for the TSL prototype

Follow a target at a bounded rate

Use a persistent output state per channel. At each sample, move toward the input by no more than rate_in_volts_per_second / sample_rate. When the remaining difference is smaller than that step, reach the input exactly. This preserves DC gain, avoids overshoot, and keeps maximum slope independent of transition size. It is a plausible functional model, not a transistor-exact reconstruction.

Control direction and present test settings

The three knobs remain normalized internally to 0.0-1.0. At 0.0/fully counterclockwise, the prototype selects 1000 V/s; at 1.0/fully clockwise, it selects 0.05 V/s. Thus clockwise adds glide, consistent with the 256 operating description. The intervening logarithmic knob map is a prototype choice.

Current comparison setting: +5 V CV shifts the speed upward by the full 20,000:1 rate span; negative CV shifts downward, with limits at 0.05 and 1000 V/s. The alternative tried was +1 V doubling speed. Neither is a verified TSL CV sensitivity. Positive CV currently reduces glide, opposite to the manual knob's glide-amount direction.

Audio and CV behave differently

A fixed DC voltage eventually passes at its original level. Audio can become much quieter when its waveform changes faster than the rate limit. For a sine wave of peak amplitude A and frequency f, peak slope is 2*pi*f*A V/s. A 100 Hz, 5 V-peak sine needs about 3142 V/s to pass without slope limiting, exceeding this prototype's 1000 V/s maximum. Fastest response is therefore not true audio bypass.

For portamento, patch sequencer pitch CV through TSL into an oscillator's pitch input. Feeding oscillator audio through TSL explores waveform shaping instead. The user's sequencer-driven 1947b test revealed this distinction.

Unverified research leads

NotesTowardsAtsl claims particular matched pairs, capacitor dielectrics, MC1458 devices, extra mode switches and drift mechanisms. The available EMS TSL sheet does not verify those claims. Do not add leakage, thermal drift, a linear/log switch or a particular component model solely from these fragments. The separate 3xSlew schematic is included as a non-EMS topology comparison, clearly labeled.

5 / Source register and appendix index

Archive page numbers below are 1-based PDF pages. Original source sheets retain their resolution and scan defects. Use the document bookmarks to jump to each sheet. The appendix is reference material, not a redraw or a newly verified build specification.

Packet pages	Source and scope
7	Computer Synthi brochure, source p. 2: published Synthi 100 TSL behavior.
8	Synthi 100 service archive, source p. 25: EMS 32/12 Issue C, TSL circuit.
9	Synthi 100 service archive, source p. 21: Random Generator with related slew shaping.
10	Synthi 100 service archive, source p. 63: sequencer system context.
11-35	Complete 25-page Synthi Sequencer 256 service collection. ADC is packet p. 29; DAC is p. 33; system diagram/slew block is p. 35.
36	3xSlew_Schematics.pdf: non-EMS comparison circuit.

Stored sources and SHA-256

EMS%20%20Synthi%20100%20Computer%20Synthi.pdf
`cca1abf921cba9ae3f254e0f2c138aa5fb4f3eb0b104114210136bf75c559094`

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ems_synthi_sequencer_256_sm.pdf
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3xSlew_Schematics.pdf
`5d11f98359029e7fa2f1f061eebf85ef8f8c87ca37aed873e1a3e4ad84266812`

Archive root: voltage-modular/Teletape Workshop/references/. Additional local leads: NotesTowardsAtsl; # Software Slew Implementation Block.py; EMS rack-lineup photograph. The code sketch supplies a difference clamp but requires an explicit sample interval to interpret MaxChange as a physical rate.

Primary 256 operating manual (PIN Electronics)
Studio Sound, May 1975: equipment survey, printed pp. 54-55

- Noise Levels
  - The signal to noise ratio of all voltage controlled amplifiers is better than 74 dB for an input signal level which produces a T.H.D. of 3 %. This ratio is constant over the upper 30 dB of the control range. In other words, as the amplifier gain is reduced, the noise is proportionally reduced as well. When the amplifier is cut off, the noise at its output is immeasurable.

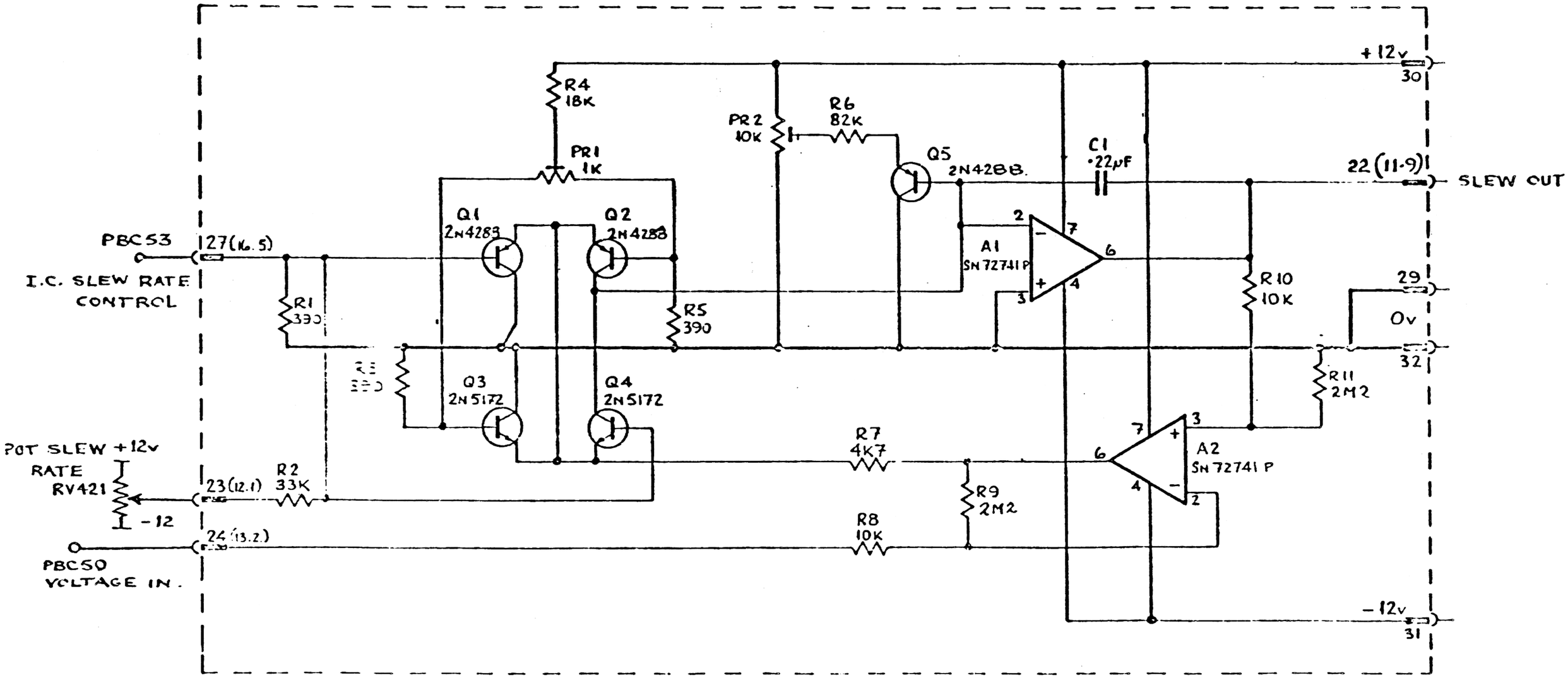
- Specification
- *Source Modules*
  - Three Voltage Controlled Audio Waveform Generators, Sine and Ramp Manual Frequency Range: Greater than 1Hz- 10kHz (extendible by voltage controls in both directions - to 0.25 Hz and 20 kHz). Sine Purity: Better than 500/0 total distortion between 10 Hz and 10KHz. Ramp Output Linearity: Departure from linearity ca. 1 % of best straight line between 10Hz and 10KHz. Voltage Control: 5v/octave. Accuracy 0.3 % departure from best straight line between 100 and 2000 KHz. Frequency Stability: Generally better than 2 % from month to month, but the oscillators usually hold their setting to within 2 cycles in a thousand during a working session. A sine shaper is included by which variable amounts of even harmonic distortion may be added.
  - Three Voltage Controlled Audio Waveform Generators, Triangle and Square. These can be varied from triangle to sawtooth ramp, and from symmetrical square to short pulse, in either polarity. Manual Frequency Range: Greater than 1 Hz to 10 KHz. Triangle Symmetry: ca.5% rise time to fall time equality. Other specifications as for sine/ramp oscillator.
  - Three Voltage Controlled Low Frequency Waveform Generators. Same details as before, but oscillators are twenty times as slow. Frequency range: Greater than 0.025 Hz (40 secs. per cycle) to 500 Hz. Voltage Control: .5v/octave. These three oscillators are intended mainly as control sources, but can be used for tone generation at the upper end of the range. All nine of the above oscillators have synchronisation inputs so that they can operate at an integral multiple of another oscillator, providing a huge variety of waveforms which can be used in additive synthesis.
  - Three Noise Generators. Variable from white (central position of colouration control) to dark or light positions (low or high pass filters). Distortion: In white position, frequency content is flat ca. 3 dB from 100 Hz- 10 KHz. We recommend several noise sources, because with different filterings more than one can be used for different purposes.
  - Dual Output Random Control Voltage Source This device produces two control voltages which move abruptly from one level to another. The distribution of levels is rectangular rather than Gaussian, and the iwo outputs are uncorrelated in level, but synchronous in time. The mean time between changes, and the variance about that mean are manually controllable. The distribution of times is rectangular, and in common with all other time controls devices in the studio, a control range of at least 1000:1 is available. Controls:
    - Amplitude Variance: Up to 2.5v symmetrically positive and negative (x2, each output separately controllable).
    - Time Range: From approximately 10 secs to 10 ms if no time variance is applied.
    - Time Variance: From equal steps to 1000:1 variance. Effect of variance control is obviously limited when time range control is at either entrance.
- *Treatment Modules*
  - Three Voltage Controlled Trapezoid Generators with Integral Envelope Shapers These devices might be described as voltage programmable four segment curve synthesizers, the output being available as two control voltages and as the modulation of an audio signal. The basic waveform produced (at each control output) has four stages:
    - 1 Delay. The output remains constant and negative for a controllable time after the generator is triggered.
    - 2 Attack. The output rises to a fixed positive value at a controlled rate.
    - 3 On. The output remains constant and positive for a controllable time after the completion of attack.
    - 4 Decay. The output falls at a controlled rate to its initial value.

    In addition, there is a second output which lags behind the first by one quarter of a complete trapezoid cycle. Thus the time set for, say, on in respect of output 1, becomes the time for attack from output 2, and so on. The amplitude and polarity of both outputs may be adjusted independently so that if they are summed (on the patchboard) any continuous four line function which ends at the value at which it starts may be produced. This arrangement gives an extraordinary flexible range of envelopes. The envelope shaper portion consists of a logarithmic voltag~ controlled amplifier permanently connected to one of the trapezoids. The overall control of each cycle of operation may be in a number of ways selected by a switch, having the following positions:
    - 1 Signal threshold. Any signal above a certain level initiates a single cycle.
    - 2 Hold sequence starts when control is positive. Decay does not start until control goes negative.
    - 3 SIngle shot positive zero crossing initiates a single cycle.
    - 4 Free run.
    - 5 Gated free run positive level allows the sequence to free run. Sequence stops at the end of a cycle when level goes negative.

    Panel Controls: Initial delay time 2 ms to 20 sec., Attack time 2 ms to 20 sec., On time 2 ms to 20 sec., Decay time 2 ms to 20 sec., (these parameters may also be voltage controlled over their entire range.) Trapezoid phase 1 output level (centre zero knob), Trapezoid phase 2 output level (centre zero knob), Signal level control, Push button (to initiate cycle), Trigger mode selector switch, Envelope shaper is logarithmic to within 3 dB over its 60 dB range., Voltage control function of time parameters is ideally exponential to within 10 % (of dependent parameter) over a range of 1000:1 Departure from ideal is gradual beyond these limits. This permits a single voltage applied to all inputs to compress the time scale.
  - Four Voltage Controlled Filter/Oscillators (Low Pass to Resonating) Operation as a sine source. Frequency Range: Greater than 5 Hz-20 KHz. Sine Purity: Better than 3 % total distortion between 10 Hz and 10 KHz. General Noise Figure for Oscillators: Spurious outputs not greater than 0.1 %. As filters, they are adjustable from Low Pass to Resonating Filters, covering the entire sonic range. Frequency Range: Greater than 5 Hz to 20 KHz. Low Pass Position: Cut off rate 12 dB for first octave and 18 dB per octave thereafter. Resonator Position: Maximum stable 'Q' factor-20. Accuracy of Exponential Voltage Control Function: ca. 1% between 100 Hz and 2000 Hz. Note: Operation as a voltage controlled oscillator is limited by the time taken to respond to an abrupt change. Maximum slew rate is about 2 ms per octave.
  - Four Voltage Controlled Filter/Oscillators (High Pass to Resonating) Similar, but complementary to Low Pass Filters.
  - One Octave Filter Bank This consists of eight resonating filters, fixed-tune one octave apart, in the range 62.5 Hz-8 KHz, seperately controllable.
  - Two Voltage Controlled Reverberation Units Each spring unit has two elements with delays of 35 and 40 ms. Maximum Reverberation Time: 2.4 seconds. Useful Frequency Range: 30 Hz-12 KHz. Voltage Control Range: ca. 2v from no reverberation to maximum reverberation.
  - Three Voltage Controlled Slew Limiters This device is a unity gain amplifier in which the output exactly follows the input at a rate whose maximum (slew) is defined by a control voltage. One application might be to interpose the device between the pitch control voltage from a keyboard and the oscillator whose pitch is to be controlled. If the key velocity voltage were then applied to the slew control input, the player could produce a glissando between any two notes that he played, the rate of glissando being controlled by his touch. Steady State Gain: 1 ca. 1 % Steady State Linearity: ca. 0.05 % (BSL) Range of Slew Control: 1 ms to 10 sec. Voltage control of slew is exponential. Note: Unlike all other devices, no output level contro; is provided as the device has unity gain.
  - Three Integral Circuit Transformerless Ring Modulators These very efficient modulators also include amplifiers, and can therefore be used in series for double or triple modulation. Maximum Input for Undistorted Output: 1.5v p-p to each input. Breakthrough with 1.5v on one input only:5mV p-p (-60 dB)
  - One 256 Event, 6 Simultaneous Parameter Digital Sequencer This machine is, in fact, a small special purpose digital computer, complete with analog to digital and digital to analog converters. It provides a sequence of control voltages which may be used on any of the devices in the studio.
    - The operation of the sequencer may most easily be described in terms of conventional music, although it must be remembered that the design by no means limits it to this kind of operation.
    - The sequencer stores 256 notes and plays each note at the correct time and for the correct duration. It simultaneously provides two voltages, one of which might be used to define pitch and the other loudness. It is capable of controlling three voices, each with duration, pitch and loudness.

B. EXTERNAL NOTES ADDED. 9/4. 23.4.71.

C R2 WAS 39K. 9/4 7.8.71.



CIRCUIT DIAGRAM TRIPLE SLEW LIMITER. DRG. Nº EMS 32/12 ISSUE C

ELECTRONIC MUSIC STUDIOS (BOURNEMOUTH) LTD.

B C2 WAS 10KP
C1 WAS 220KP
C4 WAS 1μF
Q13 WAS 2N4852
Q21 WAS 2N5163
CB WAS 10KP
R43 WAS 6K8
Q28 WAS 2N5163
C12 WAS 10KP
R56 WAS 6K8
D/S OF Q21
DID GO TO +12V
D/S OF Q28
DID GO TO +12V
MH 16.4.71
C; R68 & R69 ADDED
Q13 W/S UT46
~~MH~~ 3.12.71
& C4 WAS 2·5μF

NOISE SOURCE
SCHMIT TRIGGER
RAMP & HOLD
TIMING OSCILLATOR
+12V
R1 47K
R3 470
Q3 2N5172
R7 1K5
Q6 2N4288
R13 6K8
PR1 100K
R16 470
Q10 2N5172
PR2 100K
R22 22K
R24 22K
R25 18K
R27 22K
R29 470
Q16 2N5172
R26 10K
R2 68K
Q2 2N5172
Q1 2N5172
0V
C2 10μF
Q4 2N5172
R5 4K7
Q5 2N5172
R8 4K7
R9 220K
R12 2K2
D2 HS1395
D3 HS1395
R18 10K
20
KEY OUT
RV 363
R28 4K7
C1 10μF
-12V
D1 HS1395
Q7 2N4288
Q9 2N5163
Q8 2N4852
C3 10Kp
R4 4K7
R6 8K2
R11 8K2
R14 4K7
R15 4K7
R17 10K
R19 330
Q11 2N4288
Q12 2N4288
Q13 2N4852
Q14
Q15 2N5172
R20 27K
R21 12K
C4 1μF
PR3 100
R23 1K2
R30 2K7
14
15
18
TIME VARIANCE
RV360 10K
CW
+12V RV359 MEANTIME 10K -12V
CW
FIRST VOLTAGE
SECOND VOLTAGE
30 +12V
RAMP HOLD
SLEW LIMITED BUFFER
RAMP
HOLD
SLEW LIMITED BUFFER
D7 HS1395
R53 470
R48 22K
R69 10M
R45 1K5
D5 HS1395
R44 4K7
R43 3K3
R42 2K2
R40 470
PR4 100K
Q20 2N5172
R35 22K
R32 1K5
R69 10M
R31 4K7
27
VOLTAGE OUT PUT 2
PBC
R56 3K3
R55 2K2
PR5 100K
Q27 2N5172
R49 15K
C9 10μF
C10 10μF
D6 HS1395
R36 15K
C5 10μF
C6 10μF
D8 HS1395
0V
29
Q30 2N4238
Q28 2N5457
R46 6K2
Q23 2N4238
Q21 2N5457
Q19 2N4288
Q18 2N4288
R39 6K2
Q29 2N4352
Q26 2N4288
Q25 2N4288
Q24 2N5172
Q22 2N4852
CB 100Kp
R41 4K7
Q17 2N5172
C12 100Kp
R54 4K7
R52 4K7
R51 15K
R47 3K3
C11 47Kp
R33 4K7
R38 15K
R34 3K3
C7 47Kp
-12V
31
13
12
17
22
28
OUTPUT 1
RV 361
RANGE CONTROL LEVEL
VOLTAGE 1 RANGE
CW
VOLTAGE OUTPUT 1
PBC 84
OUTPUT 2
RV 362
RANGE CONTROL LEVEL
VOLTAGE 2 RANGE
CW

NOTE.
1 POSITION 2P ON SYNTHI 100
2 MODULE WIRING, SEE DRG E.M.S. 31/20

CIRCUIT DIAGRAM RANDOM GENERATOR DRG E.M.S 32/08 ISSUE
ELECTRONIC MUSIC STUDIOS (WOBURN MILTON) LTD

MCMURDO 16 WAY

MCMURDO 16 WAY

1 O
3 O
2,10O

6F20
6FR20

11V

6400/25V

0.25Ω W.W

2N3055/BDL30

UBA
5V. 2.6A.

8
2
1 WT. 14/.
4 BRN.14/.
3 BRN.14/.
GRN. 14/.
15,16.
7,8.

1KpF

+5V
2.6A

UBA
12V. 250 mA.

18V

8
7
2 ORG 7/.
1 ORG 7/.
4 BN 7/.
BN 7/.
14
6

1KpF

+12V
250 mA.

COMM. 0V

UBA
12V. 250mA.

18V

8
7
2 YEL. 7/.
1 YEL. 7/.
4 GY. 7/.
GY. 7/.
5
13

1KpF

-12V
250mA

UBA
5V. 400mA.

13V

8
7
2 RD 14/.
1 RD 14/.
4 BLU.14/.
3 BLU.14/.
9
12

1KpF

-5V.
400mA

Fig 3

16μF

N L
Hv
-5V.
-12V
+12V
-5V

NESTRA ELECTRONICS LTD.

31/49.

P.S.U. TYPE 3100 SK 3102/10

SYNTHI SEQUENCER 256

EMS

Electronic Music Studios London Limited,
49 Deodar Road London SW15 2NU
Telephone 01-874 0042/874 2363 Telex 928327
New York: 408 East 78th Street NY 10021 Tel: 212 734 7344
Telex 42 40 33

MODE SWITCH
Position 1: NEW KEY. A key pulse is sent at the start of each event even if no keyboard attack is made.
Position 2: NEW PITCH. Will not send a key unless there is a gap between events.
Position 3: REWRITE B, D, F: Allows rewriting of second parameters without disturbing first. See (16) SPECIAL.

CLOCK REMOTE JACKS
RESET, START and STOP clock functions can be remotely operated.

CLOCK DISPLAY
Four digit numerical display. Shows position of clock in sequence, and therefore can identify individual events.

LAYER SELECT
Decides which of the four tracks is being recorded. Others can be heard back if occupied.

EXTERNAL INPUTS
Leading to (16) and (17). A, C, E and B, D, F plus Key 1, 2, 3, 4.

INPUT SENSITIVITY
With (16) adjusts levels of different types of control input.

INPUT SOURCE SELECTOR
Four position switch selects (1) KEYBOARD (built-in KB(5)), (2) SPECIAL for rewriting B, D, F (see 21), (3) CENTRE-ZERO for EMS-type (bipolar) controls, (4) END-ZERO for Moog-type (+) controls.

EXTERNAL CLOCK IN
For operation by e.g. an oscillator (see EXT/INT CLOCK SWITCH).

CLOCK RATE OUTPUT
Voltage proportional to clock speed. For remote indication and control.

METER AND METER FUNCTION SELECTOR
Normally set to show the amount of store occupied, but switchable to log track voltages etc.

POWER SWITCH AND PILOT LAMP

EXT/INT CLOCK SWITCH
In conjunction with (EXTERNAL CLOCK IN).

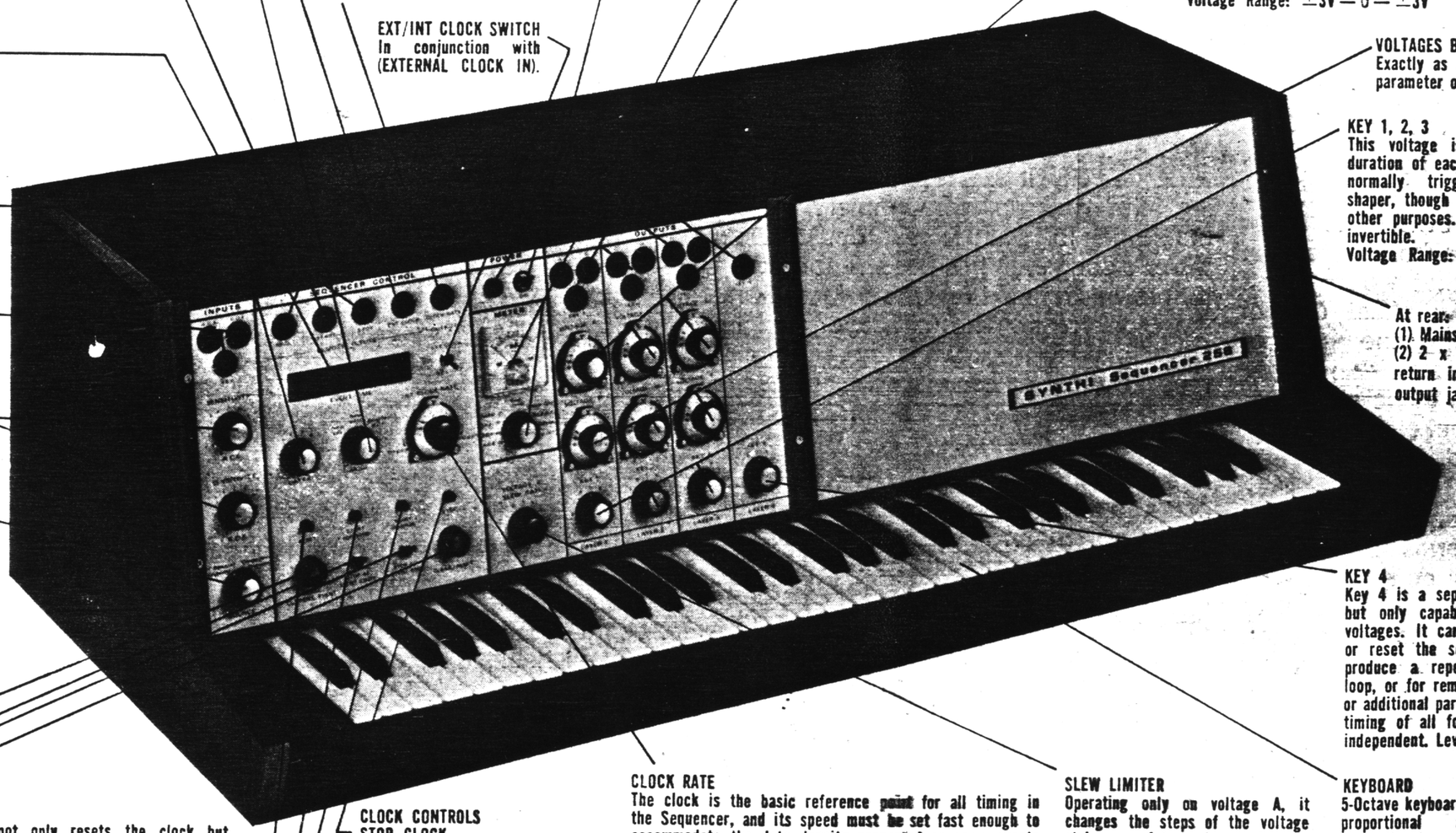
VOLTAGE OUTPUTS
All ten output voltages at these jack sockets.

VOLTAGES A, C, E
These three slow-motion controls compress, expand or invert the voltage range at the output, without changing the basic data in the memory. If this voltage were controlling an oscillator, it would define interval. A, C, E are the first parameters of each of the three layers, and the controls give zero halfway, expanding normally and inverted in each direction.
Voltage Range: ±3V — 0 — ±3V

VOLTAGES B, D, F
Exactly as A, C, E for second parameter on each track.

KEY 1, 2, 3
This voltage is present for the duration of each event, and would normally trigger an envelope shaper, though it can be used for other purposes. Also variable and invertible.
Voltage Range: ±5V — 0 — ±5V

At rear:
(1) Mains input and fuse.
(2) 2 x multi-way sockets. return input and output jacks.



KEY 4
Key 4 is a separate fourth layer, but only capable of storing key voltages. It can be used to stop or reset the sequencer clock, to produce a repeated sequence or loop, or for remote control, cueing or additional parameter control. The timing of all four tracks is quite independent. Level as Keys 1, 2, 3.

KEYBOARD
5-Octave keyboard delivering 'pitch'-proportional and dynamically proportional voltage outputs. Connected to sequencer when (16) set to KEYBOARD.

SLEW LIMITER
Operating only on voltage A, it changes the steps of the voltage staircase into slow changes. Applied to pitch, for example, it produces a portamento.
Range of Slew Rate control: 1mS — 10S.

CLOCK RATE
The clock is the basic reference point for all timing in the Sequencer, and its speed must be set fast enough to accommodate the data density presented or some events will be missed. A voltage proportional to clock speed is available at a jack output (24) for simultaneous control of any parameters as clock speed is adjusted. Alternatively a ramp or pulse oscillator may be used as an external clock, and a switch (25) disconnects the internal clock in this case.
Clock Rate Range: 0.1 — 200Hz

CLOCK CONTROLS
STOP CLOCK — clock stops counting
START REVERSE — clock goes backwards
START FORWARD — clock runs normally
RESET — return to zero without stopping clock, unless button held down.

CLEAR/RESET
This button not only resets the clock but erases the entire memory, and is guarded to prevent accidental operation.

EDITING CONTROLS
STOP AT EVENT END.
When this switch is down the clock will stop at the end of the current event, until restarted.
STOP AT EVENT START.
Clock will stop at start of next event.
ERASE CURRENT EVENT.
Pressing this button will erase a selected track while clock is running. Used with 13 and 14 and the clock controls.

SYNTHI SEQUENCER 256

EMS
Electronic Music Studios London Limited,
49 Deodar Road London SW15 2NU
Telephone 01-874 0042/874 2363 Telex 928327
New York: 408 East 78th Street NY 10021 Tel: 212 734 7344
Telex 42 40 33

The SEQUENCER 256 is, in fact a small special purpose digital computer, complete with analog-to-digital and digital-to-analog converters, exactly as incorporated in the SYNTHI 100. The ease of operation plus the freedom it allows completely outdates traditional sequencers, which require tedious setting up, cannot be properly edited, and give only short sequences. This machine provides a sequence of control voltages which can be used with any type of synthesiser available, and it was partly in response to owners of other makes of synthesiser who wanted an advanced sequencer that we made it separately available. An offset control automatically lines up the control voltages to suit Arp, Moog, or any other non-EMS make. For the same reason of flexibility, both jack sockets and multi-way plugs are provided.

The operation of the sequencer is most easily described in terms of conventional music, but it must be remembered that its design by no means limits it to this kind of operation. The information which is to be stored and subsequently reproduced is loaded in as control voltages, generally but not necessarily from a keyboard. The sequencer can store and play 256 events at the correct time and for the correct duration, providing two simultaneous voltages per event which might, for example, be used to define pitch and loudness. It can control three such parameter pairs simultaneously (overlaps are quite permissible), and the 256 events can be distributed in any desired way between the tracks. Two of the three tracks can be played back (and heard) while the other is being recorded or edited.

To record a sequence the composer first sets an electronic clock running. As he plays each note, the memory at that instant records how many clock pulses have elapsed since the start of the sequence, and how many during the time that the note was held down. It simultaneously remembers which note on the keyboard was played and with what velocity it was struck (also passing the voltage through for direct monitoring of what is recorded). When a basic sequence has been recorded on that layer, the next layer is selected and the clock reset. The first layer can be heard playing back while the second is recorded, and so on until all three layers are recorded (six parameters plus key voltages for attack).

The sequence may then be edited just like a tape recorder only with no inertia (and no razor blades and splicing tape!). The programmed material is 'fast spooled' forwards or backwards, and then advanced event by event, so that each note may be isolated prior to being modified or erased until absolute perfection is achieved. A further track, called Key 4, contains only pulse data, but can be used in a wide variety of timing and remote control applications, for automatic looping of some of the sequence, or as an extra attack pulse.

The 10,750 bit memory is of the "contents addressable" type, so called because the long (42-bit) words contain all the sorting data necessary and can be stored in the first convenient vacant space. Imagine a circular tube full of blank dominoes, the whole line of dominoes rotating at high speed past a window in the tube, so that all 256 dominoes are examined every 0.01 seconds. Data from the inputs is presented to the memory as a complete package of information arranged as follows: 13 bits for start time, 13 bits for end time, 6 bits each for the two parameters (or one 12-bit word if one high resolution parameter is preferred), and four bits for keying, routing (which layer?) and 'housekeeping'. The word drops into the first blank domino available, 'labelled' with its time information related to clock pulse numbers. The high resolution given by 26 bits for time enables all events to be referred to zero, and this is one of the really important breakthroughs achieved by the Sequencer 256 — editing of any event is possible without disturbing any others, because all events have complete individual data. Many sequencers operate by referring the current event only to the previous one, and editing is consequently impossible or at best extremely difficult. When the sequence is played back, imagine a diary with a page for each day, but with all the pages torn out and restacked in any order. This is the appearance of the full memory — 256 dominoes but in no particular order. The clock starts and the whole diary is checked at each pulse (max. time for search 0.01S). Is there anything for February 2nd? No. February 3rd? Yes. Event starts and here are the voltages. Etc., etc. When editing, a particular domino or group of dominoes can be erased, rewritten and fitted neatly back into the jumbled diary.

When used with non-EMS equipment, the inputs can be connected either through the jacks or via the multi-way sockets behind the instrument, and there is special provision to offset the control range for end-zero systems (e.g. Moog), EMS equipment having bi-polar controls.

SPECIFICATION

All the important facts are shown in the legending accompanying the numbered photograph overleaf, detailing the ten variable controls which adjust the amplitudes of the sequencer's output voltages; and the further eight knobs, six buttons, three switches, meter and digital clock which supervise the actual operation. Do not imagine from the multiplicity of controls that the 256 is difficult to operate — it is very simple. Sometimes an external connection is necessary — for example when a clock loop is required a jack cord is taken from Key 4 Output to Reset, and a pulse recorded on Key 4 at the point where you wish to reset the clock to zero automatically.

All in all, the SEQUENCER 256 represents a huge step forward in the technique of voltage control, and working with it opens up new horizons for composers and performers.

NB: The SEQUENCER 256 as fitted in the Synthi 100 is identical in operation but different in layout and some details, since it is built in. But all the functional operations can be identified easily from the panel legending and related to the photograph overleaf.

Set up controls as follows:

Envelope Shapers 1, 2 and 3
Mode; hold on, delay = 0, attack = 0, on = 0, decay = 5, Trapezoid 1 & 11 = 0, signal level = 10.

Oscillators 1, 2 and 3
Shape = 5, sine level = 10, ramp level = 0, freq. = 5.

Keyboard Controls
Upper keyboard pitch = 8.1 , keyboard velocity = 0, env. control +5, switch = down.

Sequencer outputs
Layer 1 voltage A = 5.2, voltage B = 0, key 1 = +5,
Layer 2 voltage C = 5.2, voltage D = 0, key 2 = +5,
Layer 3 voltage E = 5.2, voltage F = 0, key 3 = +5.

Output Channels
Channel 1; filter = 0, pan = -5, not off, level to 3-5,
Channel 2; filter = 0, pan = +5, not off, level to 3-5,
Channel 3; filter = 0, pan = 0, not off, level to 3-5.

Sequencer Operational Control
Set all switches to 'up' position.
Switch event distribution to layer 1.
Set clockrate vernier to 5.

Plug in the keyboard unit using the connection cord provided.

Now connect up the sequencer mains supply by inserting the labelled plug into the distribution panel previously mounted in item 9. The 'sequencer event time' should now read some arbitrary number. This is located on the furthest right hand vertical panel. If it is not showing 4 red/orange coloured digits, recheck the fitting of the 'Nestra' power unit. A further check if necessary, can be made by opening the hinged panel upon which the 'sequencer event time' display is mounted. You will now be able to see the back end of the 'Nestra' power unit, and a check of the supply voltages can be made. The values are printed on the unit, namely +12, -12, -5, +5 and 180 volts for the display.

To check sequencer, carry out the following operations:
on the 'sequencer operational control' panel;

A. Press 'Stop'.
B. Press 'Clear/Rest Memory'. The event time should now read 0000.
C. Press 'Start Forward'.
D. Play a tune on the top keyboard.
E. Press 'Reset'.
F. You should now hear your tune replayed. If necessary adjust output channel 1 level to suit, and set channels 2 and 3 to the same level. Assuming your tune replays alright;
G. Press 'Stop'.
H. Press 'Reset'.
I. Set event distribution switch to layer 2.
J. Press 'Start Forward'.
K. Play another tune on the top keyboard. Whilst doing this, you will hear your original tune being replayed.
L. Press 'Reset'. You should now hear both the tunes replayed.
M. Press 'Stop'.
N. Press 'Reset'.

O. Set event distribution switch to layer 3.
P. Press 'Start Forward'.
Q. Play a third tune on the top keyboard. This time you will hear the two previous tunes replayed.
R. Press 'Reset'. You will now hear all three tunes.

NOTE; It is possible to latch a key on, if you press 'Reset' or 'Stop' during a note. This will clear later when another event on that layer ocurrs.

If the sequencer fails to operate, check all the p.c. cards are firmly pushed home. The power supplies have already benn checked. If these are still O'K, then pull each card out in turn, and push it back in, in case it has been misaligned in transit.

Sequencer Notes

1. If the sequencer does not appear to produce a musical scale, try tuning the appropriate voltage vernier;
   Layer 1 = Vernier A
   Layer 2 = Vernier C
   Layer 3 = Vernier E
   The correct settings for these are given on the calibration sheet supplied with the machine.

2. If the keyboard gives a strange non-musical progression of notes, the ADC addressing may be wrong. Check that the keyboard pitch control is set as specified earlier; this figure is also given in the calibration sheet.

3. If 1 layer in particular does not give correct output signals, suspect its DAC (digital to analogue converter).
   Layer 1 Location C (Red Handles)
   Layer 2 Location D (Red Handles)
   Layer 3 Location E (Red Handles)

4. If event time counter does not count forward after step C (item 17) the realtime clock may not be running (located on board A, red handle). Check first that the pin specified for location vertical 51, horizontal 82 on the signals patchboard is in place. If it is, patch the clock voltage out to the oscilloscope Y1 input by putting an orange pin at vertical 56, horizontal 82. A ramp waveform of about 4 volts peak to peak should be seen.

5. Further checks will require the sequencer circuit diagrams which are supplied with the set of literature.
   Good checks are:-
   A. Look for timing signals as described on 32/44. These can be picked up on the wiring at the back of rack 3. Drawing 31/47 (of which a large copy is supplied) shows all signals and interconnections.
   B. Check +2.5v and -2.5v supplies on pins 19 and 20 of boards B, C, D and E. These voltages are generated on board B.

   There are a number of other checks that may be done if a sequencer fails to operate. If in trouble, the best solution is to write to:-

   Mr. N. Vivian,
   Technical Manager,
   Hilton Electronics (Wareham) Ltd.,
   Johns Road,
   Wareham,
   Dorset,
   England.

   with explicit instructions as to what the symptoms of the fault are,

TECHNICAL SERVICES

D.C.C.O. INDEX OF DIGITAL NAMES

CCP — Conversion (analog - Digital) Complete, 500Hz Square Wave.

CFP — Convert first parameter, 250Hz Square Wave which alternately connects two input voltages to the A.D.C.

CMB — Clear memory button. Causes recirculation control logic to write "Free" (i.e. (0)) in MRo.

CKY — Clear key flip flops. Enables clearing of key flip flops associated with the D.A.C.s. Key flip flops generate the output keying voltage to control envelope shapers.

DCP — Down Count Pulse. Real time clock pulse (1 μs) to decrement display counter.

DTo — 2MHz 1:3 pulse used to strobe comparison between each converter's digital value and time reference into switching flip flop

DT1 — Time reference 1MHz (L.S.B.)

DT2 — " " 500KHz

DT3 — " " 250KHz

DT4 — " " 125KHz

DT5 — " " 62.5KHz

DT6 — " " 31.25KHz

DT7 — 31.25KHz pulse used to reset switching flip flop

DIGITAL ANALOG CONVERTER TIME REFERENCES

EKC — Enable clearing key flip flops

EWF — Enable writing "free"

GFF — Go flip flop. Is set during the first real time clock cycle after the clock is started. When set it prevents the clock from being stopped. (by S.T.P.).

HLD — Shift timing signal which strobes certain data into latches which are held constant for part of each word cycle.

KEY4 — Output of fourth key flip flop, which when converted to a ±2.5v signal is available at patch boards for stopping or resetting the clock.

KNB — Latched signal representing the state of the key input at the start of each word cycle.

LACE — Enable loading of first parameter (A,C.E.) D. to A. converters.

LBDF — Enable loading of second parameter (B.D.F.) D.to A. converters.

LDA — Enable generation of LACE and LBDF while data is being entered.

Latched signal generated when the layer select switch is the same as the current output of the layer select bits from memory.

DRWG.6.04 INDEX OF SIGNAL NAMES

MRo — **Recirculating memory input/output.** Bit MRo is a (0) when the location is free, and becomes a (1) when complete data is entered, and an event recorded.

MR1) MR2) MR3) — Layer select bits. When MR1 is a (1) the data will be directed to D. to A. converters A. and B., when MR2 is a (1) data goes to C. and D., when MR3 is a (1) data goes to E. and F. When all three bits are (0) key 4 is set or cleared, depending on event time bits.

MR4))) MR9) — These six bits carry the data. Input is from the analog digital converter, output is to the digital analog converters. Data is in pure binary format, 00_8 representing - 2.5v, 77_8 representing +2.5v. MR9 is the least significant bit.

MR10)) MR19) — These ten bits carry the event time, i.e. the count on the real time clock at the instant that the key input was activated or released. Format is pure binary. MR19 is the least significant bit.

MS1) MS2) MS3) — Layer select switch (on console). These correspond to MR1 thru. MR3.

MS4 — Layer select switch key 4 position. This signal not used as Key 4 is defined as $\overline{MS1} \cdot \overline{MS2} \cdot \overline{MS3}$.

P1 — Shift timing signal. Defines the first half of each word. The first half contains layer select, start of event time, and first parameter.

P2 — Shift timing signal. Defines the second half of each word. Second half contains end of event time and second parameter.

PH1 — Shift timing signal. This signal drives the $\emptyset_1$ M.O.S. clock driver. Leading edge writes the first 20 bits of each word, trailing edge reads them.

PH2 — Shift timing signal. Drives $\emptyset_2$ M.O.3 clock driver. Leading edge writes the second 20 bits of each word trailing edge reads them.

STE — Stop-At-End-Of-Event Switch. As S.T.B. except that the clock is stopped at the end of the event.

STP — Stop the clock pulse, produced by S.T.E. or S.T.B. when an event to be reproduced is detected.

TEQ — Signal generated when recorded event time equals current (real time) clock time.

TLQ — T.E.Q. latched at end of first read cycle.

TOV — Thousands overflow (carry) from decimal display counter. Used to reset the 10 bit binary real time clock counter which would otherwise count to 1024.

TUN — Thousands underflow (borrow) from decimal display counter. Used to set the 10 bit binary counter to 7747_8 (=999) when the count goes backwards through zero.

DRG.6.04 INDEX OF SIGNAL NAMES

UCP — Up count pulse. Real time clock pulse when time direction is forwards. Increments display counter.

UPD — Time direction control.

W1 — Shift timing signal. Pulse generated in centre of P1, used for strobing data into D to A. converters, and for generating S.T.P.

W2 — As W1 except that it occurs during P2. Also used to clear key flip flops.

WAD — Write analog digital converter in memory. When W.A.D. is low stored data is recirculated.

WET — Write event time in memory. When W.E.T. is low stored event time is recirculated.

WMS — Write layer select switch in memory. When W.M.S. is low stored layer select is recirculated.

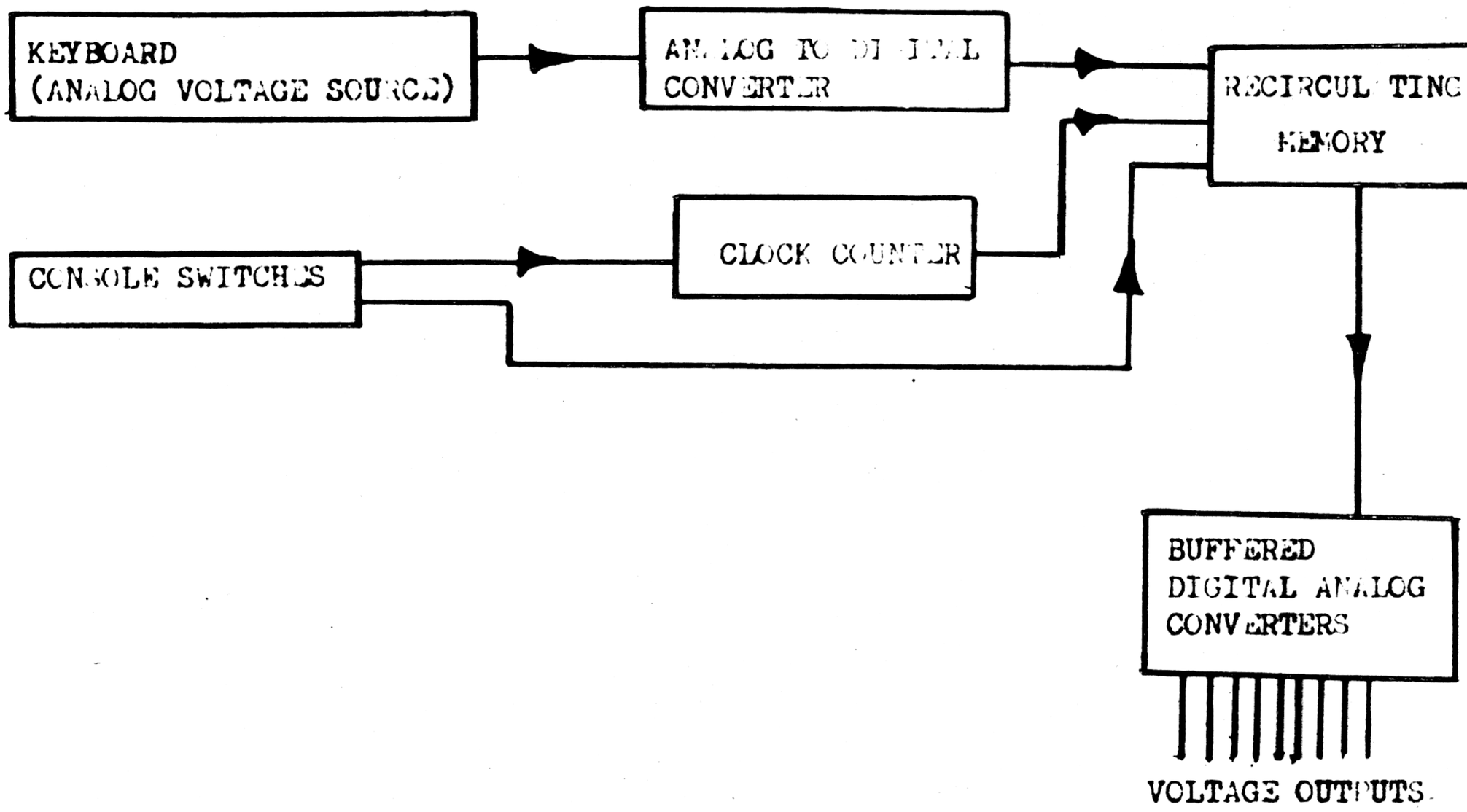
ZOB — Reset (to zero) real time clock counter and display counter. Is actually used to preset zeros into display counter (rather than activate the clear line).

SEM — Special edit mode. In this mode the second parameter data is rewritten without changing first parameter or event time.

MRLo — Is MRo latched at particular time.

SEQUENCER DESCRIPTION 6.05

BLOCK DIAGRAM



FUNCTIONAL DESCRIPTION

The sequencer is a small special purpose digital computer organised to operate rather like a multi track data recorder. Input data for each track (or layer) is fed in separately, but the outputs of all tracks are available simultaneously.

Input data is in the form of two analog voltages and a keying level which defines when data is to be stored. These signals are most easily provided by an organ type keyboard - one analog voltage being proportional to the pitch of the highest key pressed, the second proportional to the velocity with which it is struck, and the keying level remaining high as long as any key is held down.

In the normal mode of operation the user would simply play in the first line of a three part performance, then restart the sequence, and while hearing the first line, play in the second line, and so on. The machine is organised such that data may be entered while previous data is being reproduced, even if the latter is on the same track.

Various other operational modes are provided to allow selective editing of data already entered.

The design of the machine allows simple expansion of the capacity of the memory and the number of tracks.

MEMORY ORGANISATION

The memory used is in the form of long serial access dynamic shift registers. These are similar to delay lines in that data is only available at the output at a fixed time after being entered, and also that the motion of data cannot be stopped.

Twenty shift registers are used, each being of 512 bits. Data is organised in 40 bit words, each consisting of two 20 bit bytes.

Each word, (which might represent only musical note), is organised in the following way:

First Byte		Bits
Free/Used	- This bit records whether the location contains data.	1
Layer Select	- Bits defining which track the data belongs to, source of this data is switch on console	3
First parameter	- Binary value of one of the two input voltages (normally pitch).	6
Start of event time	- Content of clock counter at instant when key was pressed.	10
	TOTAL	20
Second Byte		
Second Parameter	- Binary value of the second input voltage (normally loudness).	6
End of Event Time	- Content of clock counter when key was released	10
Unassigned	- Wasted in present design.	4
	TOTAL	20
		40

Note that all the information about an event is contained in the 40 bit word. The order in which words go through the shift registers is entirely fortuitous. This allows the clock counter to be independent of and asynchronous to the shift clock, provided that a complete circulation of data occurs during each of the former's clock cycles.

This form of memory organisation is known as "Content Addressable".

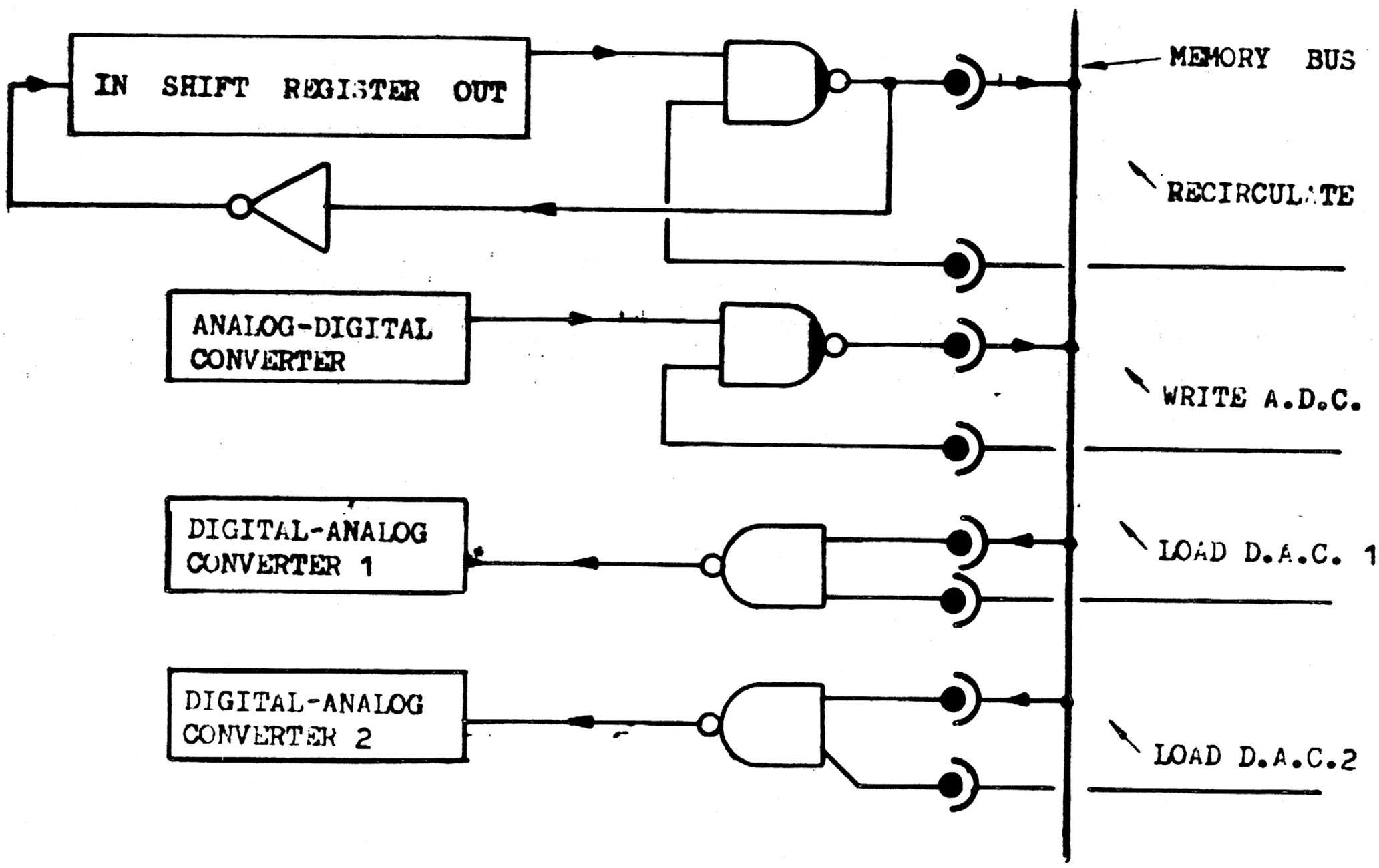
PHYSICAL ORGANISATION

The sequencer consists of some 120 multi pin packages and a smaller number of discreet components.

Layout is designed for minimum interconnections between the ten boards containing the digital functions.

All boards plug into what is basically a 60 way unibus. Certain bus lines are time shared using open collector logic. This means that a shift clock reference must always be used when examining these busses during debugging.

The bus lines that are shared are those carrying data into and out of the memory. Note that one wire in general may be part of one of four data paths at any instant depending on the phase of the shift clock.



In fact the shift registers used have two clock inputs, each dealing with alternate bits.

Start-of-Note-Time is written continously in all uncommitted locations while the key is released, except the first one after the key is released. End-of-Note-Time is written in all uncommitted locations.

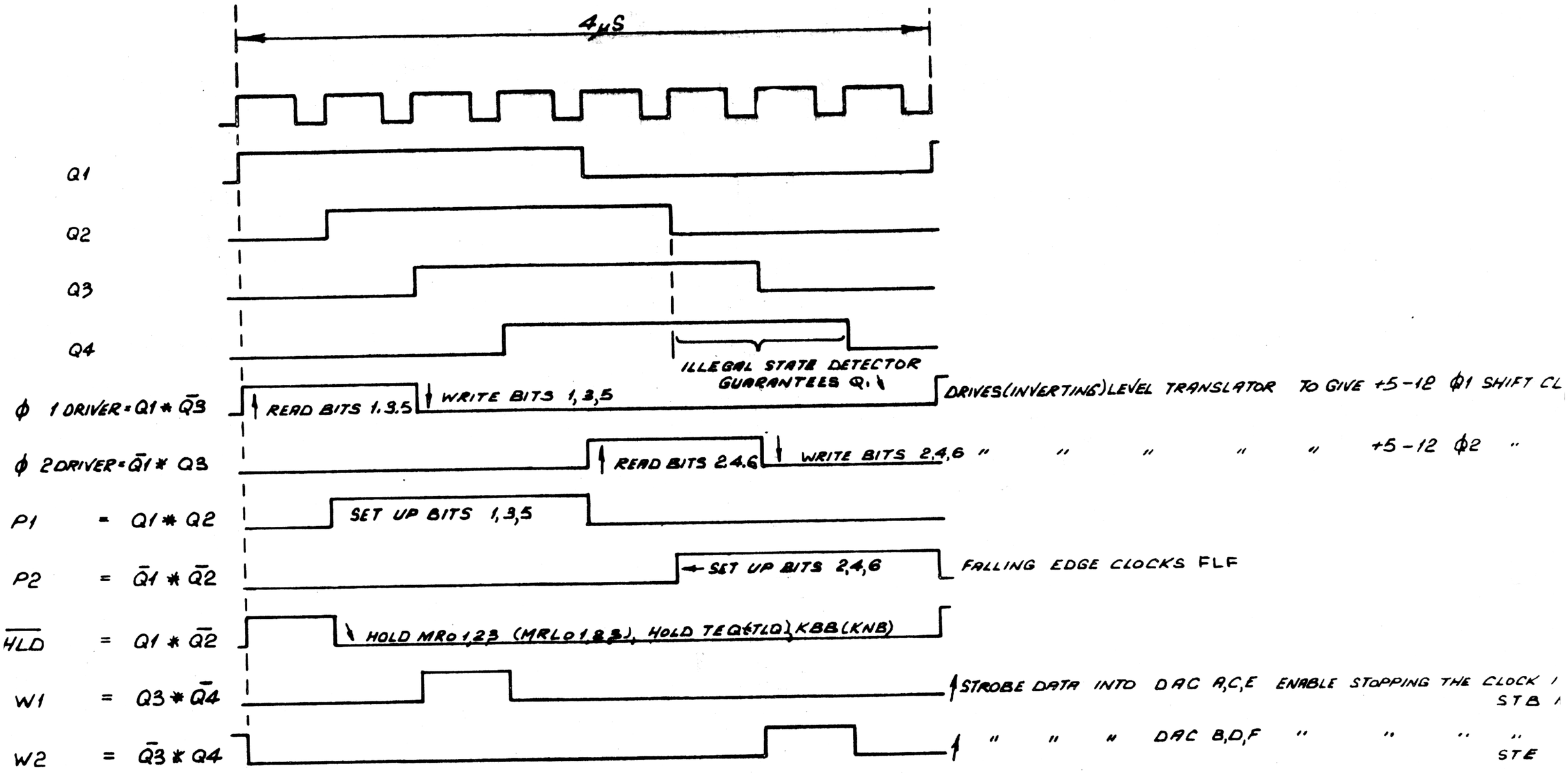
As soon as the key is released the first uncommitted location is labelled committed.

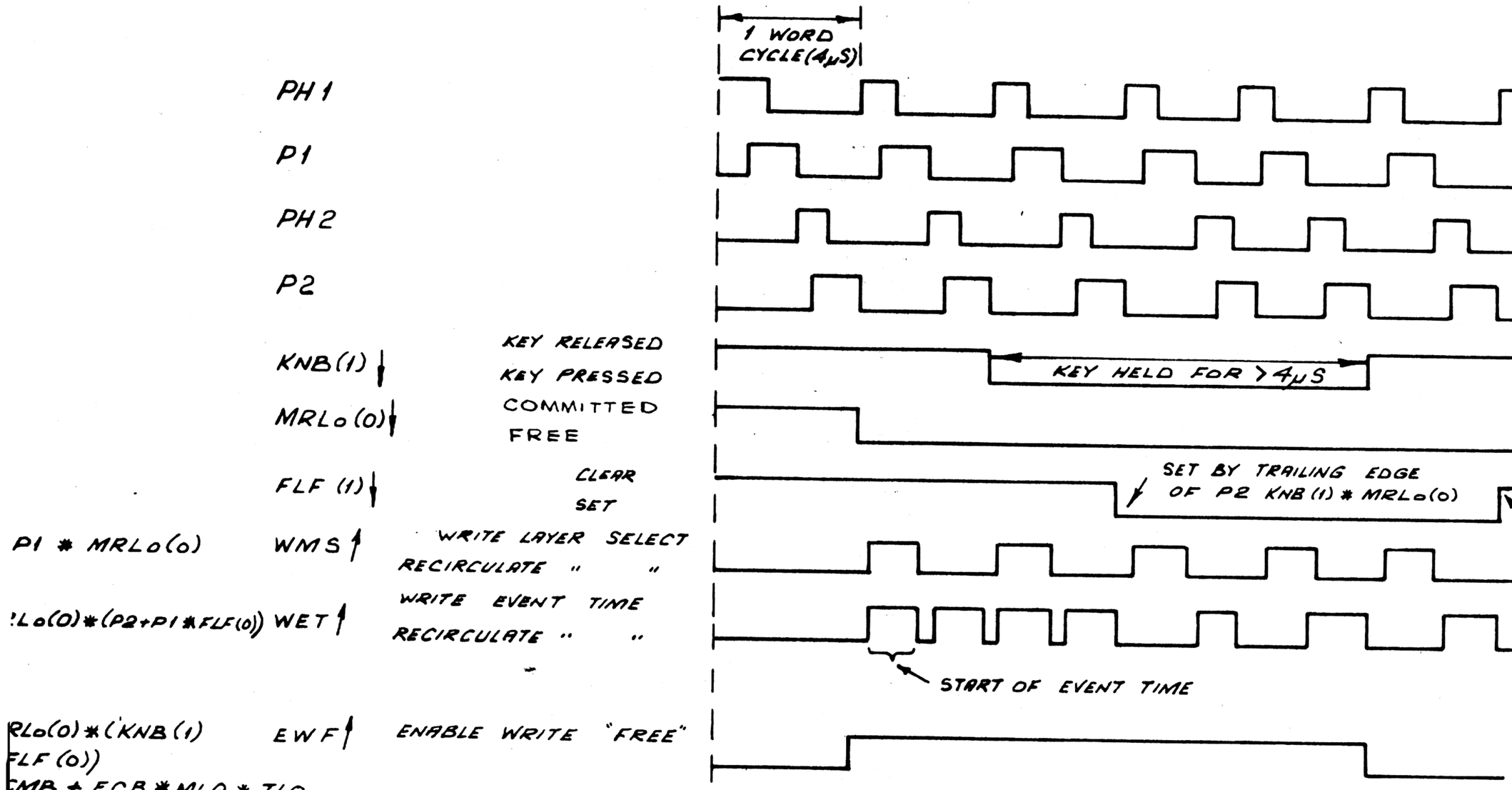
The flip flotp, F.L.F. becomes set at the end of the first uncommitted word in which the key was pressed, and cleared at the end of the first uncommitted word in which the key was released.

The digital analog converters receive data either from previously recorded words, or in real time when a key is pressed. The conditions for loading a particular D.A.C. are

EITHER:	The event (start-of-note) time of a recorded word equals the clock (T.L.Q.) and the D.A.C. is selected.
OR:	That D.A.C. is the one selected by the note distribution switch.

The key output goes high when the first parameter D.A.C. is loaded and low when the end-of-note-time recorded equals the clock. While the note is played the key output goes high, (each load D.A.C. pulse would set the key flip flop), when the key is released, the first free location.


4μS
Q1
Q2
Q3
Q4
ILLEGAL STATE DETECTOR GUARANTEES Q.
DRIVES (INVERTING) LEVEL TRANSLATOR TO GIVE +5-12 Φ1 SHIFT CL
Φ 1 DRIVER = Q1 * Q̄3
READ BITS 1,3,5
WRITE BITS 1,3,5
Φ 2 DRIVER = Q̄1 * Q3
READ BITS 2,4,6
WRITE BITS 2,4,6 " " " " " +5-12 Φ2 "
P1 = Q1 * Q2
SET UP BITS 1,3,5
P2 = Q̄1 * Q̄2
← SET UP BITS 2,4,6
FALLING EDGE CLOCKS FLF
HLD = Q1 * Q̄2
HOLD MR0 1,2,3 (MRL0 1,2,3), HOLD TEQ&TLQ) KBB (KNB)
W1 = Q3 * Q̄4
STROBE DATA INTO DAC A,C,E ENABLE STOPPING THE CLOCK / STB /
W2 = Q̄3 * Q4
" " " DAC B,D,F " " " " STE


1 WORD CYCLE (4μS)
PH1
P1
PH2
P2
KNB(1)
KEY RELEASED
KEY PRESSED
KEY HELD FOR > 4μS
MRLo(0)
COMMITTED
FREE
FLF (1)
CLEAR
SET
SET BY TRAILING EDGE OF P2 KNB(1) * MRLo(0)
CLEARED BY TRA
EDGE OF P2 IF
MRLo(0) * KNB
P1 * MRLo(0)
WMS
WRITE LAYER SELECT
RECIRCULATE " "
!Lo(0)*(P2+P1*FLF(0))
WET
WRITE EVENT TIME
RECIRCULATE " "
START OF EVENT TIME
RLo(0)*(KNB(1)
FLF(0))
EWF
ENABLE WRITE "FREE"
MB * ECB * MLQ * TLQ

D-A CONVERTERS, TIME REFERENCE

E3 SN 7493 (4 BIT RIPPLE COUNTER)

E4 SN 7493

E5 SN 7493

NOTE! ON E3, E4, & E5, PINS +5V, PIN 10-0V

DTO 2 MHz SHIFT CLOCK

SDT (= $\overline{DTO}$)

+5V

0V

C2 10mF

C3 10mF

C4 10mF

C5 10mF

C6 10mF

C7 10mF

470 Ω

VLI

E10 = SN7404

E9 = SN7404

E6 = SN7474

32 KHz

250 KHz

125 KHz

500 KHz

1 MHz

DT_0 31

CCP 41

CFP 42

DT_7 20

DT_6 32

DT_5 33

DT_4 34

DT_3 35

DT_2 36

DT_1 37

NOTE!
1. ON E6, E9 & E10 PIN 7 TO 0V, PIN 14 TO +5V

SHIFT CLOCK GENERATOR

E1 SN 7413

E2 SN 7400

R1 390 Ω

C1 1000pF ± 5%

E7 SN 7474

E8 SN 7476

E11 = SN7402

E12 = SN7402

DT_0 2 MHz TO TIME REFERENCE

NOTE! UNUSED SR & SR TO VLI

8 CLOCK PULSES 4 μs/sec

PH1 13/40

PH2 10/39

P1 10/38

P2 1/50

NLD 13/49

W1 4/48

W2 1/47

SPARE

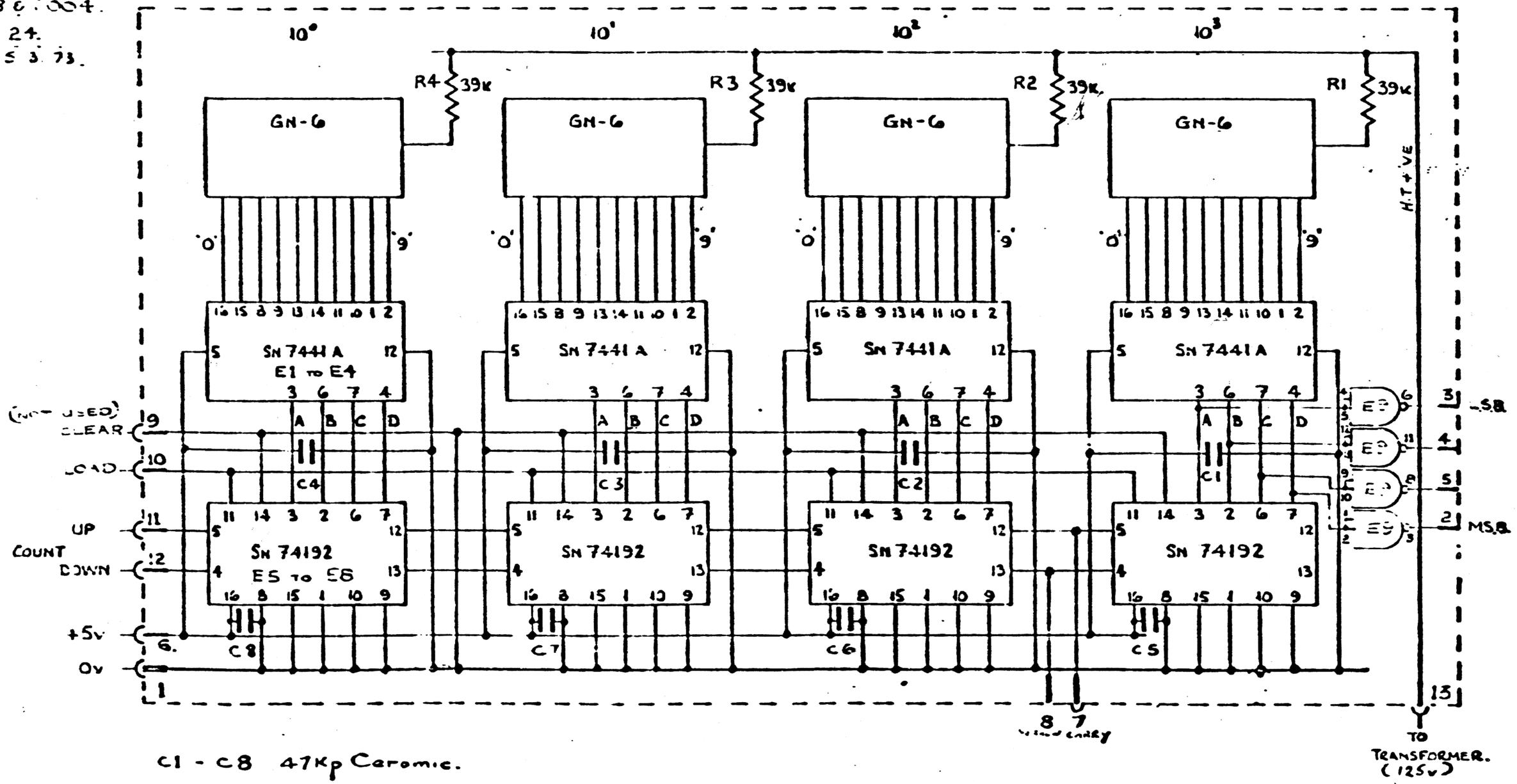
NOTE!
1. ON E1, E2, E7, E8, E11 & E12, PIN 7 TO 0V, PIN 14 TO +5V

CIRCUIT DIAGRAM FOR D-A CONVERTERS, TIME REFERENCE, AND SHIFT CLOCK GENERATOR			
ELECTRONIC MUSIC STUDIOS (Boreham Wood) LTD	ISSUE	DATE	APPD
DRG Nº 32/44	B		
		1	

DRAWN J. MASSEY

R. RIMS 330
M4 30.4.71

pins 15, 1, 10, 9 SN74192
were to ext. circ. C's added.
19.5.71.
-9 was pins pin 10 - 5
:-7. 12 8, 6-2. 8-4, 7-3.
:13-3 and term. - ack added.
2. . 1. Cap. added.
to tom. ... d form.
:038 : 004.
5 24.
5 3 73.



C1 - C8 47Kp Ceramic.

SN 7441A B.C.D to DECIMAL DECODER / DRIVER.

SN 74192 .. 4 BIT UP / DOWN COUNTER

CIRCUIT DIAGRAM 4 DECADE UP / DOWN COUNTER.

DRG. Nº EMS 32/19
ISSUE E

ELECTRONIC MUSIC STUDIOS (BOURNEMOUTH) LTD.

P1 38
MRLo (o) EKC 35
FLF(1) KNB
P2 50
MRLo (1) 23
MRLo (o) 46
KNB(1) 24
ECB 16
(ERASE CURRENT EVENT BUTTON)
MLQ TLQ 36
SEM 17
(SPECIAL EDIT MODE SWITCH)
R2 1KS R1 1KS
+5V
E4 (SN7473)
R3 1KS
+5V
FLF(1)
P1
P2
(P2)
CFP
Ecb
E1 E2 E3 E5 E6 E7 E8
(WRITE MANUAL SWITCH) 32 WMS
+5V 2 +5V
C1 C2 C3 10nF 10nF 10nF
0V 1 0V
14 0V TO CONSOLE SWITCHES
(WRITE 1ST PARAMETER)
42 CFP
CFP
(ENABLE WRITING OF R.T. CLOCK) 33 WET
+5V
R4 1KS
15 CMB (CLEAR MEMORY BUTTON)
EWF 37
(WRITE 2ND PARAMETER)
(WRITE ADC) 25 WAD
41 CCP
34 LDA

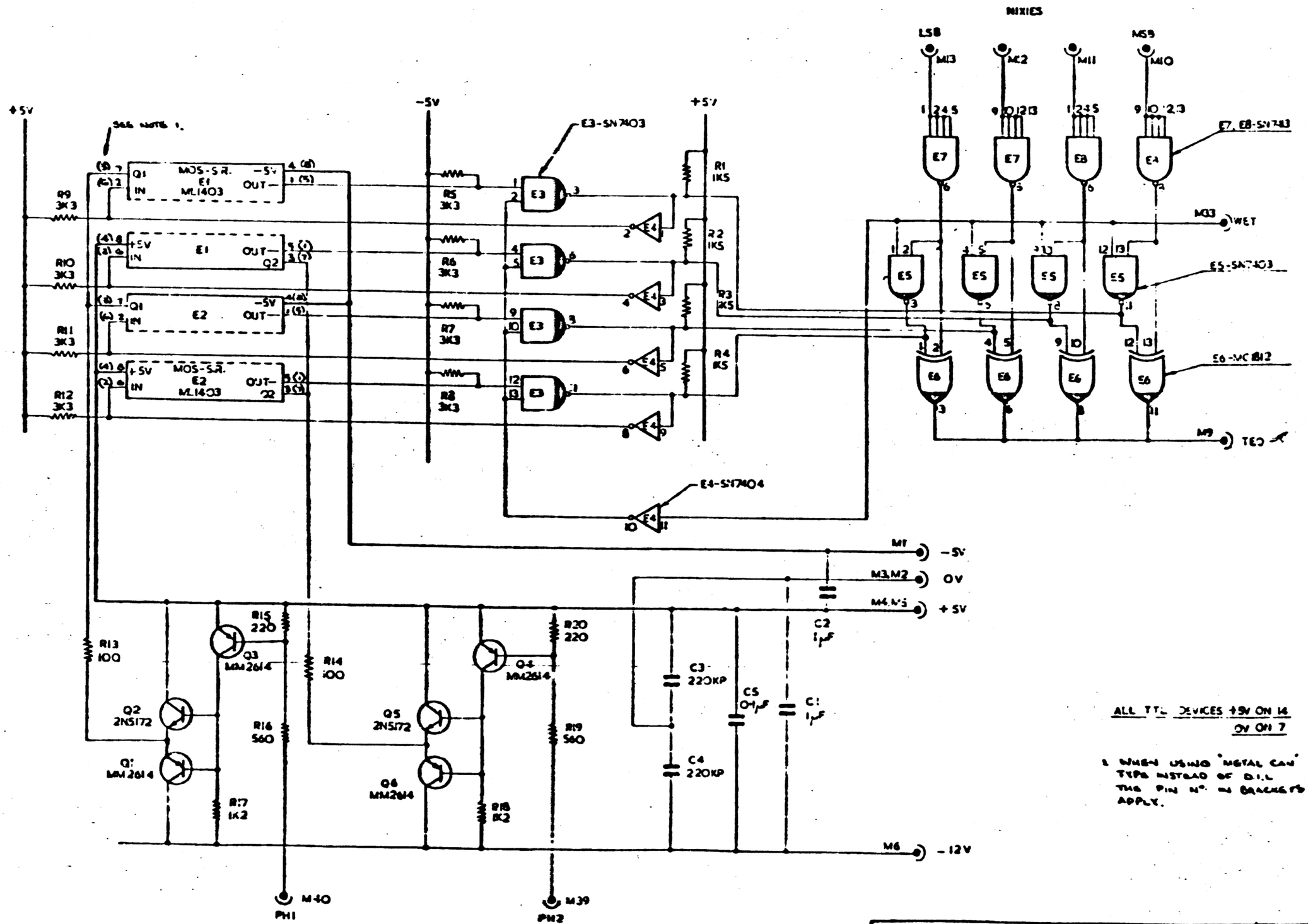
NOTES
1. E1, E2 & E8 ARE SN7402 } PIN 7 TO 0V
   E3, E5, E6 & E7 ARE SN7400 } PIN 14 TO +5V
2. ON E4 PIN 11 TO 0V, PIN 4 TO +5V

E4 (SPARE)
E2 (SPARE)
E5 (SPARE)

CIRCUIT DIAGRAM FOR RECIRCULATION CONTROL LOGIC (CARD G)

DRG No EMS 32/45	A		
ELECTRONIC MUSIC STUDIOS (BOURNEMOUTH LTD)	ISSUE	DATE	APPD

DRAWN J. MARIDEN.



ALL TTL DEVICES +5V ON 14
0V ON 7

1 WHEN USING 'METAL CAN' TYPE INSTEAD OF D.I.L. THE PIN Nº IN BRACKETS APPLY.

ELECTRONIC MUSIC STUDIOS (Bournemouth) Ltd		issue: B
MATL	TITLE X10 TIME EXTENDER	Drg Nº 32/40
FINISH	USED ON SYNTHI 100 & 256	
TOL Fine / Bar / Angles	SCALE / CHKD DATE	DRN DATE

MRL0 (0) 46 | MRL0 (0) 2) | MRL1 (1) 45 | MRL2 (1) 44 | MRL3 (1) 43

MR0(1) 50

HLD 49

MRL1(0)

SN7403

TEQ 12

SN7402

SN7400

SN74 00

EWF 37

LDA 34

MRL1(0)

SN7402

KEY4 F.F.

P1 38

MLQ 36 * TLQ

SN7400

WRITE LAYER SELECT

WMS 32

MLQ (LAYER SELECT FROM MEMORY = LAYER SELECT SWITCH)

MC1812P

MR1(1) 59

MR2(1) 58

MR3(1) 57

MS1(1) 17 LAYER SELECT SWITCH (1)

MS2(1) 16 LAYER SELECT SWITCH (2)

MS3(1) 18 LAYER SELECT SWITCH (3)

MS4(1) 14 → MS4(1) NOT USED

R1 1K5 R2 1K5 R3 1K5

+5V

24 KNB(1)

31 TEQ

Q1 2N4288

R6 4K7 0V

R7 1K5

R10 9K1

C4 220pF

10 RTP

13 KEY INPUT

R8 1K

HLD

NOTE! ON E1 TO E5 & E7 TO E11 PIN 14 TO +5V & PIN 7 TO 0V
ON E6 & E11 PIN 5 TO +5V & PIN 12 TO 0V

R4 1K5 R5 1K5

C1 10nF C2 10nF C3 10nF

+5V L

0VL

35 EKC

27 CKY

47 W2

28 LBDF

26 LACE

48 W1

29 GFF(1)

30 STP

15 STB

19 STE

+5V

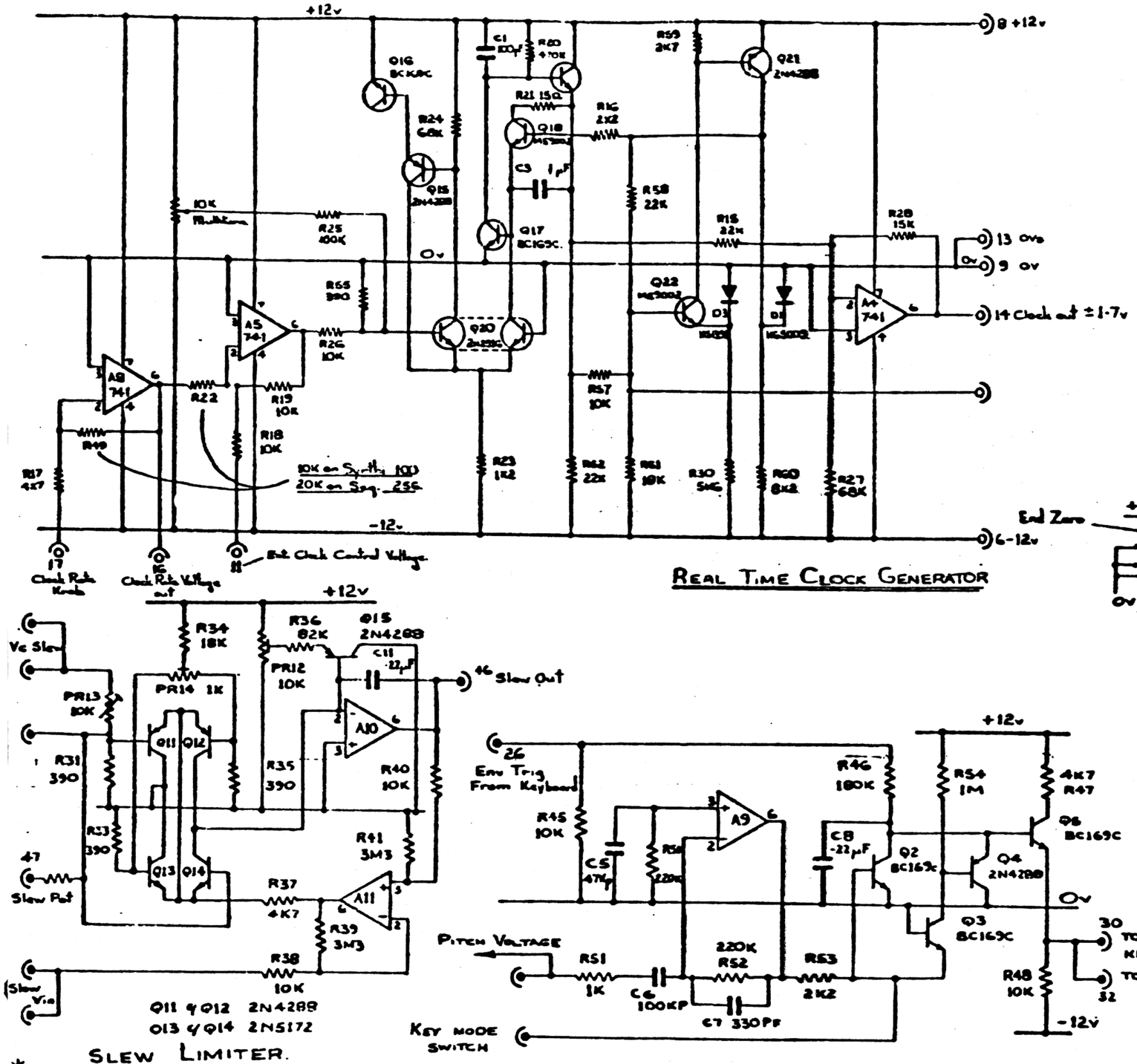
OVL

21 KEY4

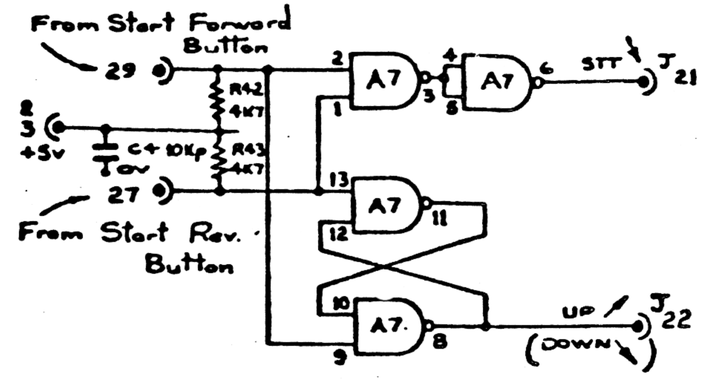
CIRCUIT DIAGRAM FOR DATA CONTROL LOGIC

(CARD H)

	B	2.8.71	K.G.
DRG Nº EMS 32/46	A		
ELECTRONIC MUSIC STUDIOS (BOURNEMOUTH) LTD	ISSUE	DATE	APP'D



REAL TIME CLOCK GENERATOR



UP/DOWN BISTABLE

Supply to A7, pin 7 GND
pin 14 +5v

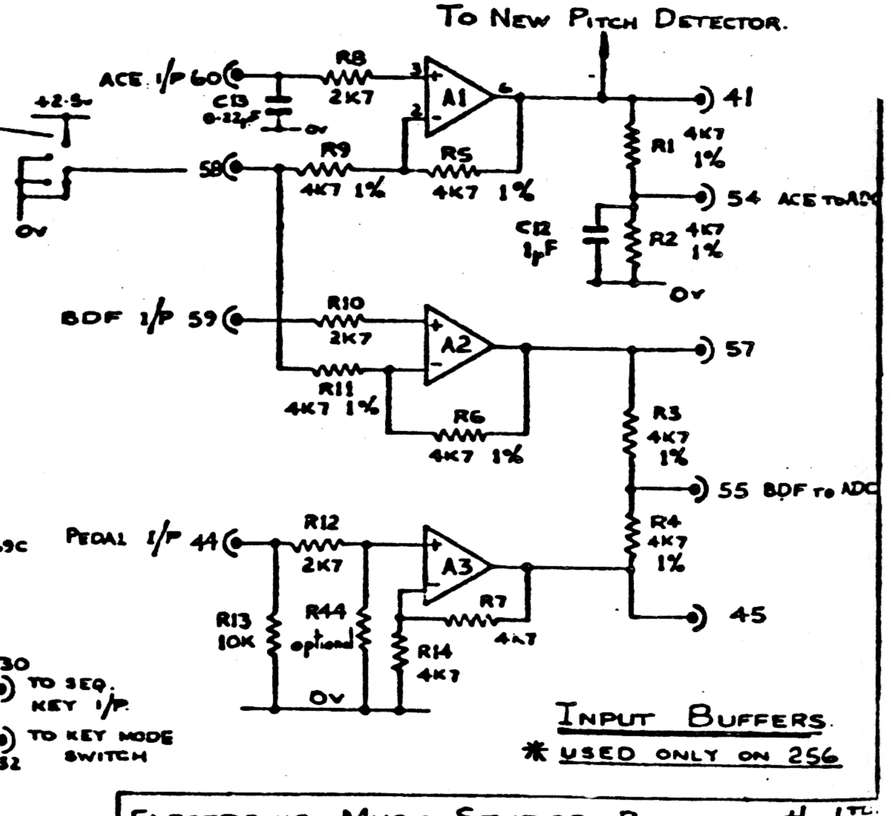
SLEW LIMITER.

* USED ONLY ON 256

A1 – A5, A8 – A11. OP AMP SN72741P or ML741C
pin 7 +12v pin 4 -12v

NEW PITCH DETECTOR.

* USED ONLY ON 256



INPUT BUFFERS.

* USED ONLY ON 256

ELECTRONIC MUSIC STUDIOS Bournemouth LTD.

TITLE AS CIRCUITS.

USED ON SYNTHI 100 & *256

32/41

DRN MJ DATE 12.8.71 ISSUE A B C D E

F.C.N.s 4873 E.C.N. 1126 C13 added 9.8.71 C.N. 21.72 C.N. A1 to B pin 13 pin 14 Shown Rev. 29.6.71

+6V
C1 10mF
C2 10mF
C3 10mF
0V
+5V
R1 470
VL1
E6 SN 7474
E9 SN 7400
G.O.B
R.T.P.
END OF R.T.P.
29 GFF(0)
24 1000's UNDER FLOW
26 UP COUNT
23 DOWN COUNT
E1 SN 74193 (4 BIT UP-DOWN BINARY COUNTER)
LOAD
UP COUNT
DOWN COUNT
CLEAR
E5 SN 74193
(NOTE! ON E1, E5 & E8 PIN 8 TO 0V, PIN 16 TO +5V)
E8 SN 74193
33 WET
NOTE! E12, E14, & E17 ARE SN7403 ON E9, U1, U2, E14, E15, E16, & E17, PIN 7 TO 0V, PIN 14 TO +5V
E14 E12 E17
R2 1K5
E17
E9 SN 7400
E9
RTP
22 UP/DOWN
R11 1K5
Z.O.B.
28 LOAD ZERO
27 1000's OVER FLOW
MR
50 49 48 47 46 45 44 43 42 41
RUN
E6
30 STP
CD
CK SD
G.O.B.
SCB
REAL TIME CLOCK COUNTER
E11 E16 E15
+5V
R16 1K6
R3 1K5
E17
31 TEQ.
9 TEQ
CLOCK CONTROL CONSOLE INTERFACE
NOTES
1. COMPONENTS NOT REFERENCED ARE SHOWN EXPLANATORY ONLY (SEE DRG:-)
32/45
32/46
1.C.3
CLEAR MEMORY
ERASE CURRENT EVENT
STOP CLOCK
RESET CLOCK
START CLOCK FWD
START CLOCK REV
G15
G16
FITTED ON PANEL
J20 R4 470
J19 R5 470
J21 R6 470
C7 10MF
C8 10MF
C9 10MF
E7 SN 7413
E10 SN 7413
E13 SN 7413
SCB
32/41 A22 up/down.
A23 A21 up/down Switch
E3 SN74121 (ONE SHOT)
C6 1mF
Z.O.B
E7
E10
E13
NOTE! >+1.2V TO TRIGGER <+0.2V TO RECOVER (BEFORE RE-TRIGGERING)
Q2 2N4288 R13 J14 REMOTE STOP
R8 10K 1K5 J13 0V (SCREEN)
Q3 2N4288 R15 J18 REMOTE RESET
R9 10K 1K5 J17 0V (SCREEN)
Vcc +5V
E3 E2
Q4 2N4288 R12 J12 REMOTE START
R10 10K 1K5 J11 0V (SCREEN)
G.O.B.
C5 1mF
C10 47KP
R7 10K
R14 1K5
J16 CLOCK IN
J15 0V (SCREEN)
Q1 2N4288
E4 SN74121 (ONE SHOT)
RTP
RTP
C4 220KP
RUN
10
C Rev Read Add J Ref
CIRCUIT DIAGRAM FOR REAL TIME CLOCK COUNTER (CARD J)
DRG E.M.S. 32/47
ELECTRONIC MUSIC STUDIOS
B 29.4.71
C 29.6.71
D 11.8.71
E 7.5.73

VOLTAGE
D.F. IN
A.C.E IN
+2·5v Gives 77
15
CFP
42
2WAY MULTIPLEXER
Q11(2N5163)
Q12 (2N5461)
R29 4m7
D3 (HS1395)
D4 (HS1395)
A12
A15
R33 4K7
R32 4K7
0V
+2·5v
-2·5v
+12v
R27 3K9
D5 (HS1395)
0V L +5v -12v +12v
1 2 6 8
0V +5v -12v +12v
+12v
R22 680
D1 (HS2062 6·2v)
D2 (HS1395)
0V
R26 1K 2%
PR2 100
R25 1K 2%
A13
C1 10 µF
C2 10 µF
-2·5v 0V +2·5v
19 9 20
R23 2K7 2%
PR1 100
A11
R24 1K (2%)
A5 A4 A3 A2 A1
A10 A9 A8 A7 A6
Q5 Q4 Q3 Q2 Q1
Bc169c
Q10 Q9 Q8 Q7 Q6
2N4288
R15 12K
R10 4K7 (1%)
R9 4K7 (1%)
R14 12K
R8 4K7 (1%)
R7 4K7 (1%)
R13 12K
R6 4K7 (1%)
R5 4K7 (1%)
R12 12K
R4 4K7 (2%)
R3 4K7 (2%)
R11 12K
R2 4K7 (2%)
R1 4K7 (2%)
R21 5K6
R20 5K6
R19 5K6
R18 5K6
R17 5K6
R16 5K6
E2 E2 E2 E2 E1 E1
SN7403
MSB
56 MR4(1)
55 MR5(1)
64 MR6(1)
53 MR7(1)
52 MR8(1)
51 MR9(1)
KEY 4 O/P BUFFER
R30 10K
Q13 (2N4288)
A14
R31 10K
-2·5v
21 KEY4 IN (TTL)
13 KEY4 OUT (± 2·5v)
write ADC. in memory WAD
NOTES
1. A1 TO A15 ARE SN72741P
PIN 4 TO -12v, PIN 7 TO +12v
NOTE!
ON E1 & E2
PIN 7 TO 0V
PIN 14 TO +5v

B Drg Corrected MH 20.4.71

CIRCUIT DIAGRAM FOR
6 BIT ANALOG DIGITAL CONVERTER
(RIPPLE THROUGH REMAINDER TYPE)
(CARD 'B')

DRG Nº EMS 32/42
ELECTRONIC MUSIC STUDIOS (BOURNEMOUTH LTD)

ISSUE	DATE	APP'D
B		

NOTE!

1 ON E2,4,6,8 & 10 PIN 8 TO -5V, PIN 4 TO +5V

ON E1,3,5,7 & 9, PIN 14 TO +5V, PIN 7 TO 0V

2 E1, E5 & E9 ARE SN7403.

E3 & E7 ARE SN7404

NOTE
Motor.

Bolt Interference

Metal Con

Function
OUT
IN2
Q2
-5v
OUT
IN1
Q1
+5v

+5V

-5V

+5V

R1 3K
R2 3K
R3 3K
R4 3K
R5 3K
R6 3K
R7 3K
R8 3K
R9 3K
R10 3K

E2 ½ DUAL (MP1403) 512 BIT S.I.S.O SHIFT REGISTER
E4 (TYPE AS FOR E2)
E6 (TYPE AS FOR E2)
E8 (TYPE AS FOR E2)
E10 (TYPE AS FOR E2)

R11 3K
R12 3K
R13 3K
R14 3K
R16 3K
R16 3K
R17 3K
R18 3K
R19 3K
R20 3K

E1, E5, E9
E3, E7

R21 1K5
R22 1K5
R23 1K5
R24 1K5
R25 1K5
R26 1K5
R27 1K5
R28 1K5
R29 1K5
R30 1K5

K33 WET
K50 MR0
K49 MR1
K48 MR2
K47 MR3
K46 MR4
K45 MR5
K44 MR6
K43 MR7
K42 MR8
K41 MR9

R40 47.
R39 47
Q4 R36 100 R37 270 MM2614
Q1 R34 100 R33 270 MM2614
Q5 Q6 MM2614
Q2 Q3 MM2614
C1 .22μF
C2 .22μF
R36 1K2 ¼W
R35 1K2 ¼W
R32 1K2 ¼W
R31 1K2 ¼W
C2 1μF
C4 1μF
C6 0.1.

K39 PHE
K40 PHI
K1 0V
K2 +5V
K4 -5V
K6 -12V

B C2.C5. R39 R40 ADDED

CIRCUIT DIAGRAM FOR RECIRCULATING MEMORY EVENT TIME BITS (CARD K)			
DRG Nº E.M.S 32/48	C	30.3.71	

NOTE! E10 = DM8002 or SN7402
E11 = SN7402

NOTE:- E2, E6 & E7 = MC1812P

MR4 (1) 56
MR5 (1) 55
MR6 (1) 54
MR7 (1) 53
MR8 (1) 52
MR9 (1) 51
LACE 26
LBDF 28
CKY 27
NRL1 (1) 45
NRL2 (1) 44
NRL3 (1) 43

E11 E10 E9

M.S.B.

QUAD LATCH E1 (DM8550 OR SN7475)

E8 (DM8550 OR SN7475)

SPARE E9

E9 = DM8002 OR SN7402

END OF NOTE

E4 (DM8550 OR SN7475)

M.S.B.

E2 E6 E7

STROBE DIGITAL COMPARITOR

(SAME PARTS AS COUNT PULSE BUT SHORTER)

LEVEL SHIFTER REF: SWITCH

R1 1KΩ
R2 6KΩ
D1 HS2036
Q1 2N4288
Q2 2N4288
Q3 BC169C
+5V
+2·5V
-2·5V
PAUS LATER FOR BIGGER NUMBERS
RESET

Low Pass Filter

R3 20K
R4 20K
C1 15nF
C2 10nF
R14 33K
A1

14 A.C.E. FIRST PARAMETER OUT

C12 0·1μF

6 -12V
20 +2·5V
13 0V
19 -2·5V

C13 47KP
C3 10nF
C4 10nF

E3 E3 E3 = SN7400

Q4 2N4288
R5 10K
R6 10K

R11 4K7
A2
R12 4K7 Omit in Synths 100
17 KEY OUT

+12V
R7 1KΩ
Q5 2N4288
R8 6KΩ
D2 HS2036
-12V

Q6 2N4288
Q7 BC169C
R9 20K
R10 20K
C9 15nF
C10 10nF
R13 33K
A3
C11 0·1μF

16 B.P.F. SECOND PARAMETER OUT
15 0V
10 0V

2 +5V
C5 10nF
C6 10nF
C7 10nF
C8 10nF
1 0V

E5 E5 E5 = SN7400

31 DT0
30 DT7
8
DT

LINKS 1 2 3

NOTE!
REMOVE LINKS 2&3 FOR LAYER 1
REMOVE LINKS 1&3 FOR LAYER 2
REMOVE LINKS 1&2 FOR LAYER 3

C11, C12 = 0·1μF
R11, R13 = 4K7
10K = 1%
30.4.71

CIRCUIT DIAGRAM FOR
D to A CONVERTERS 3 IDENTICAL
(CARDS C, D, E)

DRG Nº 32/43

ISSUE	DATE	APP'D
C	5.3.73	
B	30.4.71	

ELECTRONIC MUSIC STUDIOS BOURNEMOUTH

DRAWN J. MARSDEN

NOTE!

1. ON E2,4,6,8 & 10 PIN 8 TO -5V, PIN 4 TO +5V
   ON E1,3,5,7 & 9 PIN 14 TO +5V, PIN 7 TO 0V

2 E1, E5 & E9 ARE SN7403.
E3 & E7 ARE SN7404

NOTE
MP1403

8 BIT SHIFT REGISTER
Metal Can.

1	OUT
2	Q2
3	-5V
4	OUT
5	IN 1
6	Q1
7	+5V

B.C2 C54 R39 R40 ADDED IN 10-71.

CIRCUIT DIAGRAM FOR RECIRCULATING MEMORY DATA BITS (CARD L)			
DRG Nº E.M.S 32/49	C	30.3.71	A.9
ELECTRONIC MUSIC STUDIOS (Southern) Ltd.	ISSUE	DATE	
DRAWN J. MACDONALD	B	30.4.71	

B20 +2.5v RS1D INPUT SELECTOR

ACE INPUT 0 J1 & IS1

BDF INPUT 0 J3 & IS2

KEY INPUT 0 J2 & IS3

RS1A RS1B RS1C

RV1 ACE LEVEL

RV2 BDF LEVEL

OS18 TEST POINT

ACE INPUT BUFFER

BDF INPUT BUFFER

PEDAL BUFFER

PEDAL SET POINT

RS4/1 (METER)

RS4/2 (METER)

RS4/4 (METER)

RS5A KEYBOARD SELECTOR

RS5B

RS5C

FREE STORE

RS4/5 (METER)

NEW PITCH DETECTOR

VOLTAGE STABILIZER

CLOCK GENERATOR

CLOCK RATE CONTROL

RS4/6 (METER)

EXT. CLOCK INPUT J7 & IS8

CK RATE (↓) INPUT J8 & IS9

KEYBOARD INTERFACE

KEYBOARD

LADDER BUS W. PAIR

LATCH BUS WIPER

LADDER BUS LOW END

LADDER BUS HIGH END

LATCH BUS CONTROL

A CLOCK GEN. & LOGIC

B 6 BIT ADC

C 6 BIT DAC LAYER 1

D 6 BIT DAC LAYER 2

E 6 BIT DAC LAYER 3

G RECIRCULATION CONTROL LOGIC

H DATA CONTROL LOGIC

J REALTIME CLOCKING CONTROL

STT FWD

STT REV

ALL C.W.

RV4 VOLTAGE A SLEW RATE

SLEW LIMITER

J9 OS1 A OUTPUT

J11 OS2 B OUTPUT

J10 OS3 K1 OUTPUT

J12 OS4 C OUTPUT

J14 OS5 D OUTPUT

J13 OS6 K2 OUTPUT

J15 OS7 E OUTPUT

J17 F OUTPUT

J16 OS9 K3 OUTPUT

J19 OS10 K4 OUTPUT

RS2A LAYER SELECT SWITCH

RS4/7 (METER)

RS4/8 (METER)

RS4/7 (METER)

S4 MAINS ON

F1

N1

NESTRA P.S.U.

+5v +12v -12v -5v 0v HV+ HV- E

4 DECADE UP DOWN COUNTER

METER CCT.

RS4

10KR

1mA

-5v

ACE I/P

BDF I/P

KEY I/P

PEDAL I/P

FREE STORE

CK RATE

ACE O/P

BDF O/P

KEY O/P

REFERS TO 0V CONNECTIONS

D. ON 195.

C FREERUN WAS ON Z 8D.

& CK RATE I/P 2 WAS O/P

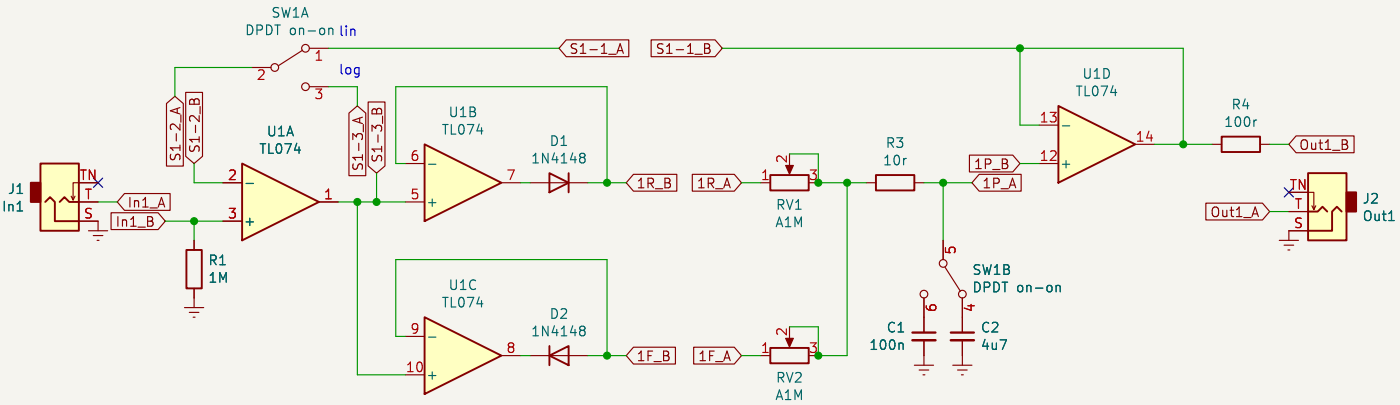
HILTON ELECTRONICS (WAREHAM) LTD

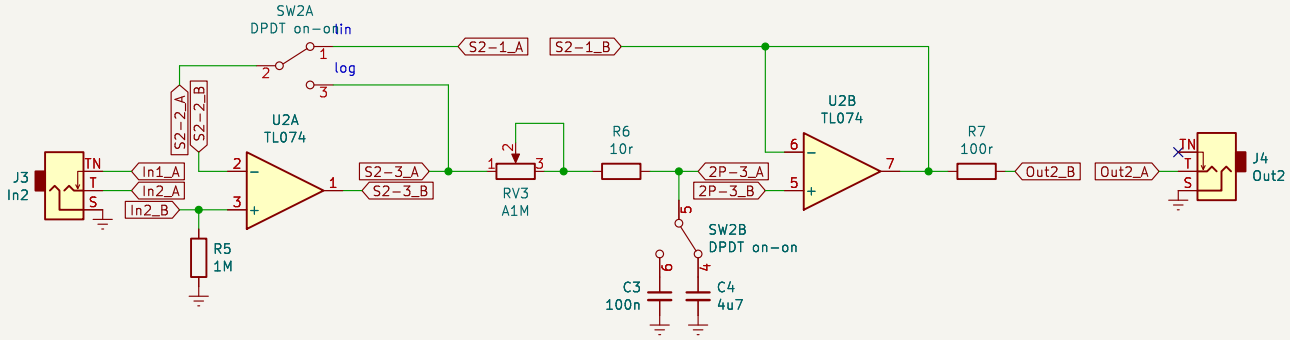
TITLE SEQUENCER 256

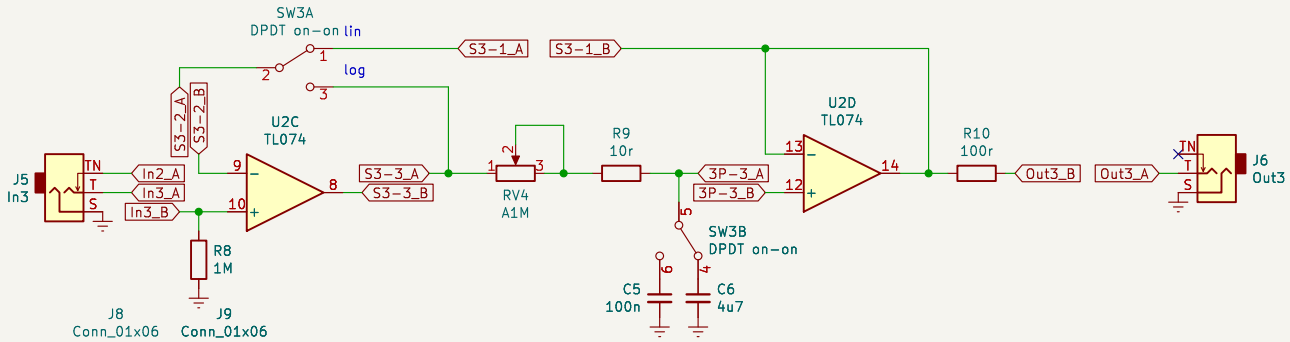
SYSTEM SCHEMATIC

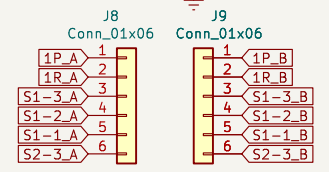
ISSUE	DRN	DATE	MAT L
A	NDV	NOV 73	
B	NDV	JAN 74	
C	NDV	2.7.74	
D	NDV	3.4.74	FINISH

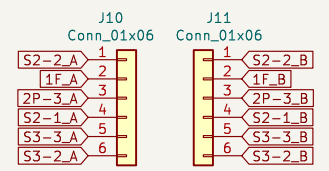
DRG NO 61 - 27

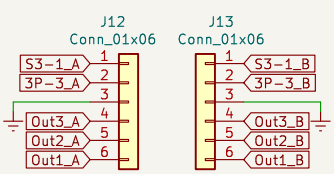


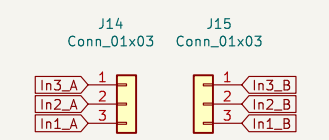


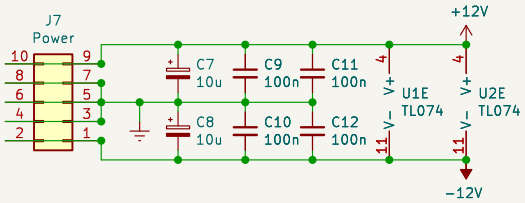












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File: 3xSlew.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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